PROTEIN BINDING STUDIES OF DIMETHYL DISULFIDE WITH ALBUMIN

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by

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DEDICATION

I dedicate this work to God, who has blessed me with strength and wisdom throughout this journey.

To my beloved parents, my rock and my biggest cheerleaders, whose unwavering love, endless sacrifices, and constant encouragement have made this dream possible. You believed in me even when I didn't believe in myself.

To my late paternal grandparents, my Dadu and my Dadi, whose blessings and values continue to guide me.

To my maternal grandmother, my Naani, for her love and encouragement, and most especially to my late maternal grandfather, my Naanu, whose dreams have become my reality.

"You have a right to perform your prescribed duty, but you are not entitled to the fruits of action. Never consider yourself to be the cause of the results of your activities, and never be attached to not doing your duty"

Bhagavad Gita 2.47

ABSTRACT

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Cyanide poisoning requires rapid therapeutic intervention, with current antidotes limited by intravenous administration challenges in emergency settings. Dimethyl trisulfide, a garlic-derived compound, represents a promising intramuscular cyanide antidote with enhanced field deployability. Dimethyl trisulfide undergoes rapid conversion to dimethyl disulfide through sulfur donation to cyanide, with ~50% conversion observed in swine studies. Understanding dimethyl disulfide protein binding characteristics is essential for predicting pharmacokinetic behavior and therapeutic efficacy. However, dimethyl disulfide volatility (boiling point 109-110 °C) presents significant analytical challenges for traditional protein binding assays requiring extended equilibration periods.

This investigation developed and validated HPLC methodology for DMDS quantification, achieving excellent linearity (R^2 = 0.9974) across 25-700 µg/mL using dimethyl trisulfide as internal standard. Vapor Transfer equilibrium studies revealed analytical challenges exceeding expected binding effects, necessitating systematic investigation of error sources.

Container integrity assessment using acetic acid demonstrated excellent seal performance (CV% = 2.1%), eliminating container leakage as the primary error source. Partitioning studies revealed complex dimethyl disulfide interactions between solution, vapor phase, and container surfaces. Acetonitrile extraction recovered surface-associated material (~85% mass balance), confirming reversible partitioning rather than irreversible

loss. Statistical analysis quantified significant surface partitioning losses (25% over 270 minutes, $R^2 = 0.64$, $p < 0.001$).

Emulsion stability characterization using laser light scattering demonstrated predictable equilibration behavior in dimethyl disulfide formulations. Analysis confirmed concentration-dependent optical signatures ($p = 4 \times 10^{-21}$) distinguishing formulation levels, with systematic scattering decline over seven days indicating micellar restructuring and surface partitioning.

Vapor Transfer equilibrium experiments showed no detectable differences in dimethyl disulfide concentration between control samples (without albumin) and albumin-containing samples, indicating negligible dimethyl disulfide-albumin binding under the tested conditions. However, documented analytical variability exceeded reasonable binding signal magnitude, preventing meaningful interpretation. Minimal transfer rates (≤0.01 dimethyl disulfide /dimethyl trisulfide ratios) suggest longer equilibration times may be required.

This investigation establishes a systematic approach to analytical challenges, providing validated dimethyl disulfide quantification methodology while identifying emulsion stability, surface partitioning, and equilibration kinetics as factors requiring control in protein binding studies. The foundation supports continued dimethyl trisulfide development as a field-deployable cyanide antidote and contributes methodologies for drug research.

KEYWORDS: Dimethyl disulfide (DMDS), Dimethyl Trisulfide (DMTS), Protein binding, Equilibria analysis, Surface partitioning, Cyanide poisoning, Equilibrium dialysis (ED)

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CHAPTER I

Background and Problem Statement

Cyanide (CN), known for its toxic properties, has been involved in mass killings, suicides, and as a tool of warfare. In 1978, members of the People's Temple in Jonestown, Guyana, committed mass suicide by consuming a fruit-flavored drink (Kool-Aid) mixed with potassium cyanide (KCN). During World War II, the Nazis used CN as an agent of genocide in gas chambers. CN is a good candidate as a chemical warfare agent. In the past, it was used as a chemical warfare agent because of its low cost of production, wide abundancy, and high toxicity.[1,2] CN can be recognized in various spheres in nature such as cosmos, lightning and volcanic eruption. Cassava roots, apricot seeds, maize, yams, and apple seeds are some of the natural CN sources. CN is also used in various industries such as gold and silver extraction industry, electroplating, many plastics and synthetic rubber industry and battery industry.[3,4] Unintentional exposures to CN are also possible, as in smoke inhalation in fires in enclosed spaces or in occupations in which CN is used in industrial processes.[1,5]

CN causes intracellular hypoxia by reversibly binding to cytochrome oxidase a_3 at the mitochondrial electron transport chain, thereby inhibiting cellular oxygen utilization and halting ATP production. This blockade disrupts oxidative phosphorylation, forcing cells into anaerobic metabolism and leading to lactic acidosis. Depending on the dose, signs and symptoms of CN poisoning may occur within seconds after inhalation and within minutes after ingestion. Early manifestations include anxiety, headache, dizziness, blurred vision, and mydriasis.[1]

Current Antidotes and Limitations

Current therapeutic strategies focus on reducing free CN concentrations through complex formation or enhanced non-toxic excretion, but these approaches remain constrained by delivery route limitations and accessibility in emergency situations.[6] The major types of CN antidotes are the scavengers (methemoglobin formers such as nitrites, cyanocobalamin) and the detoxification agents such as sulfur donors. Nithiodote™ and Cyanokit® are the two CN antidotes currently available in USA.[3] Since the available CN antidotes are administered intravenously (IV), and considering the rapid progression of CN toxicity, there is a critical need to develop an intramuscular (IM) injector kit with high antidotal efficacy to ensure timely and effective treatment of CN intoxication.

Given that currently available CN antidotes are administered in IV, the inherent limitations of IV drug delivery highlight the urgent need for a more practical and field-deployable alternative. Because IV administration requires trained personnel, reliable venous access, and poses risks such as immediate adverse reactions, infiltration, or tissue damage, it presents significant challenges in pre-hospital or mass casualty scenarios. In contrast, IM injections provide a rapid, uniform onset of action, bypasses first-pass metabolism, and is far more feasible for emergency use by first responders. This has led to growing interest in developing IM-based CN antidotes that combine efficacy with logistical ease.[7,8]

Dimethyl Trisulfide (DMTS) as Potential CN Antidote

DMTS is being investigated as CN antidote that can be delivered IM. DMTS is a natural garlic compound that is used as a food additive. Allicin gives the characteristic garlic odor and can be extracted with water or ethanol. Allicin can undergo different

reactions to produce various di- and polysulfide compounds.[3] Sulfur containing compounds in garlic, such as DMTS and diallyl disulfide, have been investigated as CN antidotes.[5] DMTS acts mainly as a sulfur donor, but it can also accelerate the formation of methemoglobin.[4,9]

For IM administration, the injection volumes need to be minimized.[10] DMTS has sufficiently low water solubility that injection volumes would be unacceptably large in the absence of formulating agents that enhance its solubility. A wide variety of cyclodextrins, cosolvents, and surfactants were tested at different concentrations for their ability to increase the solubility of DMTS. In that study, of the tested surfactants, polysorbate 80 increased the solubility the most.[9] A very substantial 735-fold solubility increase compared to water solubility was seen when using 20% polysorbate 80 as the solubilizer. DMTS was initially tested *in vivo* using polysorbate 80 as the solvent system, which enabled administration of high doses due to reduced injection volume. However, efforts to advance this formulation were discontinued because the injection volume required for scaled-up human use was too large to be practical.[10]

Protein Binding and Transport

After IM injection of an antidotal DMTS formulation, the systemic distribution of DMTS to other parts of the body takes place through blood.[11] The pharmacological effect of any drug is dependent on both its pharmacokinetic and pharmacodynamic properties, making protein binding characterization essential for understanding drug behavior *in vivo*.[12] Plasma proteins play a key role in this process. Drugs that bind plasma proteins equilibrate between two forms: free and bound.[11] Plasma protein binding affects both the pharmacodynamics and pharmacokinetics of drugs. Typically, only the free drug fraction

is biologically active. This concept, known as the 'free drug hypothesis,' establishes that free, unbound drug concentrations correlate best with pharmacological response, as only unbound drug can interact with target sites to elicit therapeutic effects.[12] Beyond mere receptor binding, unbound drug concentrations also govern tissue distribution, clearance, and ultimately, the duration and intensity of therapeutic and toxic effects. Understanding protein binding is crucial because if a drug readily crosses the blood-brain barrier by passive diffusion, the steady-state unbound brain-to-plasma partition coefficient Kpuu, brain, ss (representing the ratio of unbound drug concentration in brain tissue to unbound drug concentration in plasma at steady state) should be approximately 1, while values below 1 indicate limited access due to the action of efflux transporters.[13]

Contemporary pharmaceutical development increasingly requires comprehensive ADME (Absorption, Distribution, Metabolism, and Excretion) characterization, as understanding these parameters has become highly significant in drug discovery and regulatory submissions.[14] Patient populations with altered pathophysiology demonstrate significant pharmacokinetic variability, particularly in clearance and volume of distribution parameters, which directly impact the relationship between administered dose and achieved drug exposure, making protein binding assessment even more critical for therapeutic optimization.[15] Among the various methodologies available for protein binding assessment, equilibrium dialysis (ED) has long been established as the gold standard due to its low potential for experimental artifacts and ability to evaluate binding to the complete spectrum of serum proteins, making it the most reliable reference method for validating alternative approaches.[16]

Despite ongoing debate in the literature regarding the clinical significance of protein binding changes, regulatory agencies, pharmaceutical developers, and clinicians continue to prioritize protein binding studies due to concerns about drug-drug interactions and disease states that may alter free drug fractions and potentially impact therapeutic outcomes.[14] The integration of pharmacokinetic and pharmacodynamic principles has established clear concentration-effect relationships for many therapeutic agents, providing the scientific foundation for using protein binding data to predict and optimize clinical outcomes through rational dose adjustment strategies.[17] Recent technological advances have addressed traditional limitations of protein binding assays through the development of 96-well rapid equilibrium dialysis (RED) systems, which maintain the gold standard reliability while significantly improving throughput and reducing sample volume requirements essential for modern drug discovery applications.[16] Modern membrane technology enables relatively rapid equilibration, typically within 2-3 hours, which is particularly advantageous when studying volatile compounds where extended exposure times could lead to analyte loss.[18] Successful implementation of modern TDM programs requires not only robust analytical methodologies capable of rapid, accurate concentration determination but also clear correlation between measured concentrations and clinical outcomes, highlighting the interdisciplinary nature of contemporary therapeutic monitoring.[17]

For compounds targeting the central nervous system, protein binding considerations become even more complex due to the restrictive influence of the blood-brain barrier, which represents an additional dose determinant that must be carefully evaluated alongside traditional pharmacokinetic parameters.[19] Highly bound drugs have

long elimination half-life and do not require a sustained release dosage form, need larger doses to reach effective concentrations, and they often have limited distribution to their action and elimination sites.[11,20] When drugs bind to proteins, it helps transport them, especially lipophilic molecules, to different tissues and organs. The drug-protein complex acts as a reservoir, slowly releasing the drug to maintain a steady plasma concentration.[11] The extent of protein binding, along with the ability to cross biological membranes, are important determinants of how extensively a drug distributes throughout the body, directly affecting both therapeutic efficacy and duration of action.[12] High-permeability compounds demonstrate enhanced renal and biliary reabsorption, facilitating multiple encounters with metabolic enzymes and affecting overall pharmacokinetic profiles.[14]

The most abundant plasma protein is albumin. Albumin concentrations in blood range from 3.4 - 4.7 g/dL, constitute about 60% of total plasma protein, and are responsible for 75–80% of the plasma colloid osmotic pressure.[21,22] Human serum albumin is synthesized by liver hepatocytes and continuously secreted into the circulation.[23] Human serum albumin consists of 584 amino acids, and bears a net negative charge at physiological pH of 7.4.[24] Several variants in the amino acid sequence, alloalbumins, coexist with normal albumin. Domains II and III of human serum albumin contain two primary drug binding sites, known as Sudlow's site I and site II.[25,26]

Given albumin's critical role in drug binding and transport, understanding how DMTS and its metabolites bind to albumin can help us to understand both the pharmacodynamics and pharmacokinetics of CN detoxification by DMTS. While changes in protein binding can influence individual pharmacokinetic parameters such as volume of distribution and clearance, the most clinically relevant measure is drug exposure (area

under the curve, AUC), which directly relates to therapeutic effect and is less dependent on binding changes than previously assumed.[14]

Dimethyl Disulfide (DMDS) Metabolite and Research Gap

When a DMTS formulation is injected and transported to a CN compromised organ such as the brain or the heart, the primary detoxification reaction (Figure 1) involves the donation of a sulfur atom from DMTS to CN^- (cyanide anion) to form DMDS and SCN^- (thiocyanate). Bebarta et al. found that administering 10 mg/kg of DMTS to swine resulted in approximately 50% conversion of DMTS to DMDS in the blood, leading to an estimated effective DMDS metabolite load of 5 mg/kg.[27]

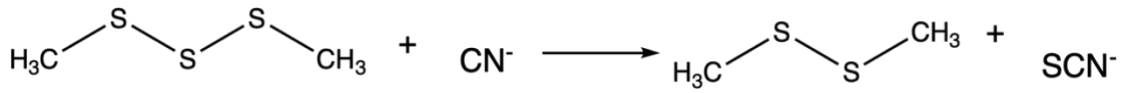


Figure 1. DMTS-CN detoxification mechanism producing DMDS metabolite. Antidotal reaction of DMTS and CN^- to produce DMDS and SCN^-.

In addition to being an *in vivo* metabolite of DMTS, DMDS is a natural component of the global sulfur cycle and is detected as a metabolite in numerous biological processes.[28] DMDS is a new pre-plant soil fumigant being developed by Arkema on a worldwide basis for the treatment of nematodes, weeds, and soil-borne plant pathogens. Although DMDS exhibits low mammalian toxicity compared to other soil fumigants, it is volatile and exposure can cause eye, nasal, and upper respiratory tract irritation, skin irritation, nausea, dizziness, headache, and fatigue.[27] DMDS has a boiling point of 109-110 °C, a density of 1.062 g/mL at 20 °C, and a pungent garlic-like odor, resembling that of decaying fish or decomposing materials.[29] Traditional ED methods assume instantaneous

protein binding equilibration and require extended incubation periods to reach steady-state conditions. The analytical challenges with volatile compounds are well-documented, with detection limits for volatile organic compounds often requiring specialized techniques to achieve ng L^{-1} sensitivity levels, which are more than 10 times lower than conventional static methods can provide.[30] However, these assumptions become problematic for volatile compounds where evaporative losses during the equilibration period can significantly compromise assay accuracy and reliability.[31] These extended incubation periods are particularly problematic for volatile compounds like DMDS. Low molecular weight volatile sulfur compounds are known to be particularly susceptible to adsorption onto metal and glass surfaces, and special precautions must be taken to prepare and maintain accurate standards. This susceptibility to surface adsorption, combined with evaporative losses during extended equilibration periods, can significantly compromise assay accuracy for volatile sulfur compounds.[32] Solid-phase Microextraction (SPME) methodologies demonstrate particular value for highly sorptive volatile compounds by eliminating losses to storage containers and transfer lines through *in situ* sampling, addressing the surface adsorption challenges that complicate volatile sulfur compound analysis.[32]

However, little is known about the binding behavior of DMDS, the primary *in vivo* metabolite of DMTS, to plasma proteins such as albumin. Since only the non-protein-bound fraction of an antibiotic is microbiologically active, and plasma protein binding significantly influences a compound's distribution, elimination, and therapeutic efficacy. Drug-protein binding interactions significantly influence pharmacokinetic behavior, as only the unbound drug fraction remains available for distribution to target tissues and elimination processes, directly impacting therapeutic efficacy and duration of action.[33]

Additionally, protein binding can be altered by disease states commonly seen in emergency situations where CN antidotes would be used, such as burns, trauma, or renal failure, making baseline binding characterization even more critical.[12] Understanding how DMDS interacts with albumin is essential for evaluating the pharmacological profile and optimizing the design of sulfur based CN antidotes such as DMTS. However, the presence of surfactants in DMDS formulations presents analytical challenges for protein binding studies. This challenge reflects common issues in volatile compound analysis, where headspace methods are preferred due to vapor-phase partitioning that complicates direct solution measurements.[34] Gas chromatographic analysis requires that all sample constituents be sufficiently volatile to move through the column system, as low volatility compounds may be retained and continue to elute during subsequent analyses, while nonvolatile compounds can condense at the column head and degrade chromatographic performance.[35] Headspace sampling fundamentally relies on vapor-phase extraction principles, where analytes partition between the sample matrix and the vapor phase above it.

SPME techniques have been developed to address volatile compound analysis challenges by eliminating solvents entirely and reducing extraction times to minutes, while providing well-defined cylindrical surface geometry that facilitates efficient extraction and desorption of volatile analytes.[32] This partitioning behavior becomes particularly complex with volatile compounds like DMDS, where achieving reproducible phase equilibrium presents significant analytical challenges.[36] As Chen et al. (2023) note, 'it is a fact that almost all tasks of surfactants involve interacting with macromolecules,' which can interfere with drug-protein binding measurements.[37] Traditional ED methods face

additional complications when dealing with surfactant-containing systems. Equilibration across cellophane membranes can occur in salt-free and polar hydrocarbon-free detergent solutions, but significant increases in equilibration times are often observed when surfactants are present. This interference with membrane transport can confound protein binding measurements by altering the assumption of rapid equilibration.[38]

The analytical characterization of DMDS-protein interactions presents unique challenges, as there is currently no standardization for pharmacodynamic models that account for the impact of protein binding of antimicrobial agents *in vitro*.[39] These challenges are compounded by the presence of formulating agents in pharmaceutical preparations. Long-chain alcohols and amine additives can markedly increase the time necessary for equilibration, and changes in ionic strength such as the addition of KCl can further prolong transport across cellophane membranes. Such interference effects can mask true drug-protein binding interactions in traditional assays.[38]

Research Rationale and Approach

Understanding DMDS-albumin binding is critical for characterizing the pharmacokinetics of the detoxification process and improving our ability to predict drug behavior in systemic circulation. Protein binding data directly impacts therapeutic dosing strategies. While the free drug bioavailability (BA) influences the likelihood of a drug candidate being approved for central nervous system (CNS) use, the interplay between BA and the drug's intrinsic potency must always be considered. This is particularly relevant for DMDS, as low permeation can sometimes be offset by high potency if only very low therapeutic concentrations are needed, they may still be achievable despite poor CNS BA.[13]

The clinical relevance of DMDS protein binding studies is needed because (1) DMDS represents approximately 50% of circulating material following DMTS administration based on swine pharmacokinetic studies, (2) DMDS may compete with DMTS for albumin binding sites, potentially altering DMTS pharmacokinetics and therapeutic efficacy, and (3) DMDS elimination kinetics could influence overall antidote duration of action in emergency scenarios. However, the analytical challenges posed by DMDS volatility (boiling point 109-110 °C) require careful validation that developed methods can achieve the precision necessary for meaningful protein binding measurements. This work represents a first step in characterizing these interactions, with the understanding that method optimization may be required before definitive binding studies can be completed.

Systematic analysis reveals that protein binding changes are clinically significant only in limited scenarios: high extraction ratio drugs administered IV or specific cases involving rapid pharmacokinetic-pharmacodynamic equilibration. For most therapeutic agents, including those cleared hepatically with low extraction ratios, changes in protein binding do not necessitate dosing adjustments.[14] The significance of protein binding cannot be generalized across different compounds, requiring careful analysis on a drug-by-drug basis, particularly for novel therapeutic agents like DMTS and its metabolites.[12]

Clinical applications demonstrate that protein binding changes have direct impacts on patient outcomes across multiple populations. In pediatric patients, the median unbound cefazolin fraction is higher in neonates than in adults, requiring assessment of cefazolin protein binding for optimization of neonatal dosing. Similarly, vancomycin shows higher unbound fractions in neonates compared to children and adults, necessitating integration

of protein binding data to achieve dosing optimization. In elderly populations, vancomycin half-life and treatment outcome are significantly influenced by severe hypoalbuminemia, with area under curve over minimum inhibitory concentration values greater than 450 mg*h/mL associated with nephrotoxicity in patients with severe hypoalbuminemia.[40] Regulatory agencies increasingly require comprehensive ADME characterization, as understanding the Absorption, Distribution, Metabolism, and Excretion profile of drug candidates has become highly significant in drug discovery. For CNS-active compounds, the most biorelevant parameter is often the unbound brain-to-plasma partition coefficient (Kpuu,brain,ss), making protein binding studies essential for regulatory submissions.[13]

This research will help clarify how the DMTS metabolite is transported, how long it remains active, and whether protein binding serves as a reservoir that modulates its BA. Advanced analytical approaches must consider not only total drug exposure but also the time-dependent concentration profiles and minimal effective concentrations required for therapeutic activity, particularly challenging for volatile compounds where concentration maintenance over time is critical.[31] This challenge is particularly pronounced in critically ill patients, where hypoalbuminemia from increased capillary permeability and decreased hepatic synthesis dramatically alters the unbound fraction of protein-bound drugs, making traditional dosing approaches inadequate and necessitating more sophisticated monitoring strategies.[15] The contemporary pharmaceutical landscape, characterized by high-speed chemical synthesis, computational ADME modeling, and early compound evaluation strategies, has created an unprecedented demand for rapid, reliable protein binding determinations that can keep pace with modern drug discovery timelines.[16] Protein binding measurements remain essential in drug development for scaling pharmacokinetic

parameters from animal models to humans, calculating appropriate first-in-human doses, and meeting regulatory requirements for comprehensive ADME characterization of new molecular entities.[14]

To address this gap, the study will employ two complementary techniques: Vapor Transfer Equilibrium (VTE) and RED. RED is an improved version of the traditionally used ED method, offering faster equilibration and higher throughput while maintaining accuracy in unbound drug quantification. While the determination of Kpuu brain ss by brain micro dialysis is considered the gold standard, as it is the only method that allows direct measurement of *in vivo* free drug concentration in the brain, this technique is time consuming and requires highly trained personnel. This limitation further supports the need for alternative methodologies like VTE for volatile compounds such as DMDS.[13] Clinical evidence demonstrates that systematic therapeutic drug monitoring programs can improve patient outcomes, with studies showing reduced hospitalization times and decreased toxicity when active TDM strategies are employed compared to traditional dosing approaches.[17]

For CNS-active compounds like antidotes, the relationship between protein binding and tissue exposure becomes particularly critical, as brain-to-plasma exposure ratios should theoretically be predicted by the relative ratio of free fractions in these matrices under passive equilibrium conditions, providing a framework for understanding compound distribution and optimizing therapeutic efficacy.[19] For narrow therapeutic index drugs like DMTS-based antidotes, understanding protein binding becomes particularly important when therapeutic drug monitoring is employed, as most routine clinical assays measure total rather than unbound drug concentrations, potentially leading to misinterpretation of

drug levels in patients with altered protein binding.[14] This precision medicine approach through therapeutic drug monitoring represents the evolution from one-size-fits-all dosing to personalized therapy, where individual patient pharmacokinetic variability is accounted for to optimize both efficacy and safety outcomes.[15]

Understanding whether a compound follows passive distribution or is subject to active transport mechanisms is crucial for therapeutic optimization, as compounds following passive distribution show excellent correlation between protein binding and tissue exposure ratios, while those subject to efflux transport may require different dosing strategies to achieve effective tissue concentrations.[19] Computational pharmacokinetic modeling has become integral to modern drug dosing optimization, enabling simulation-based approaches that incorporate protein binding data to predict optimal dosing regimens for individual patient populations, representing a sophisticated evolution from traditional empirical dosing methods.[15] VTE, developed and refined in the Thompson and Petrikovics labs, leverages the volatility of DMDS to measure binding in a two-chamber system where vapor-phase diffusion allows separation from surfactant or protein interference.[41]

This separation is critical because surfactant-protein interactions can form complex structures that interfere with binding measurements. Chen et al. (2023) describe how 'alkyl chains form the core and head groups associating with unfolded protein act as the shell' in surfactant-protein complexes, which could confound traditional binding assays.[37] This technique allows for high flexibility in drug concentrations and is particularly effective for evaluating volatile drug-protein interactions.

RED, on the other hand, uses a dialysis membrane between donor and acceptor chambers to measure equilibrium distribution of the drug in the presence of proteins and

formulating agents. While RED provides quantitative protein binding data, VTE enables analysis in the absence of interfering agents offering a holistic view of DMDS binding dynamics.[42,43] The choice of analytical model significantly impacts the interpretation of protein binding data, with different approaches yielding substantially different predictions. This emphasizes the importance of selecting appropriate methodologies that accurately reflect the physicochemical properties and behavior of volatile compounds like DMDS.[31]

Traditional well-stirred models often fail to accurately capture the complex dynamics of protein-drug interactions, particularly for volatile compounds where simple equilibrium assumptions may not hold. More sophisticated approaches are needed to account for the unique challenges posed by compounds that undergo significant evaporative losses during measurement.[31] Together, these approaches will generate critical binding parameters and improve our understanding of DMDS pharmacokinetics in the context of CN detoxification. Direct measurement of protein binding parameters is essential rather than relying on literature estimates, as binding properties can vary significantly between experimental conditions and formulations.[12]

Methodological Approaches

The goal of the proposed studies is to characterize the binding equilibria of the CN metabolite DMDS to the blood plasma protein albumin. The two primary methods that will be used for this investigation are VTE and RED. When the drug is volatile, these two methods provide complementary ways for evaluating key factors like major binding proteins, as well as the impact of protein concentration, drug concentration, and pH on the protein-bound drug fraction. This is particularly important for approving new topical drugs

and analyzing fluids with low protein concentrations, as it enables early comparisons of *in vitro* data with *in vivo* results during drug development.[41]

VTE methods have been recently developed and employed in the Thompson and Petrikovics labs to study the binding equilibria of small volatile molecules to a protein, while isolating the protein from interactions of the formulating surfactant. VTE on a two-chamber design with a formulation chamber and the protein chamber (Figure 2), separated by a nonpermeable wall and connected by a common headspace. Volatile drugs like DMDS equilibrate between the two chambers by volatilizing and diffusing through the vapor phase, while nonvolatile surfactant and protein remain in place in their respective chambers. The inherent volatility of DMDS becomes an asset in the VTE system, facilitating drug binding through vapor diffusion. Because the drug can be added in formulated form, a wide concentration range of the drug is accessible to study.

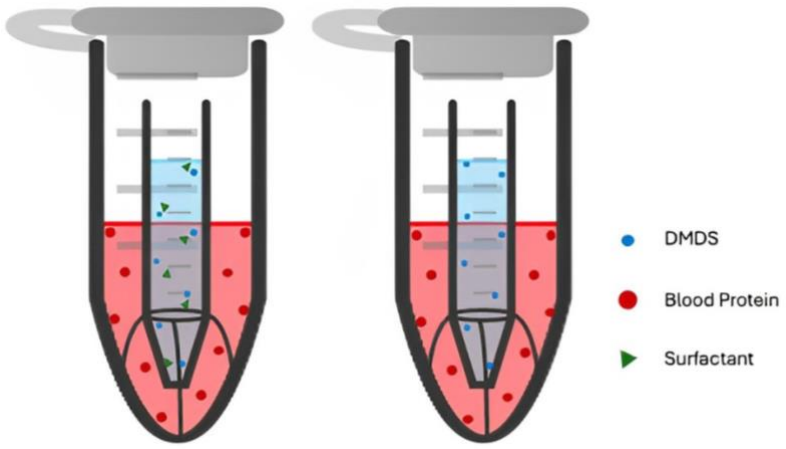


Figure 2. VTE design for volatile compound-protein binding. VTE with and without surfactant showing two-chamber system with formulation chamber and protein chamber separated by nonpermeable wall and connected by common headspace.

ED methods are used to study the binding equilibria of small drug molecules to a protein in the presence of any formulating agents. Like VTE, ED relies on a two-chamber design. However, in an ED apparatus, the donor and acceptor chambers (Figure 2), are separated by a dialysis membrane that is permeable to drug molecules and formulating agents, but impermeable to proteins. Traditional ED methods require an equilibration period of 6 to 24 hours.[42,43] RED techniques speed up analyses by enabling multiplexing of the experiment and by increasing the surface to volume ratio.[42] Although it is possible to carry out an RED experiment in the absence of surfactant, as shown in Figure 3. VTE methods described earlier overcome this limitation. Alternative analytical approaches for volatile compounds face their own limitations. SPME techniques for volatile sulfur compounds have shown that even small differences in the composition of standard and sample can result in large errors in quantitation, highlighting the need for innovative methodological approaches.

The VTE methodology addresses these challenges by utilizing the inherent volatility of compounds like DMDS as an analytical advantage rather than a limitation.[32] While ED remains the preferred method for protein binding studies due to its automation compatibility and straightforward implementation, these established protocols become particularly challenging when applied to volatile compounds.[44]

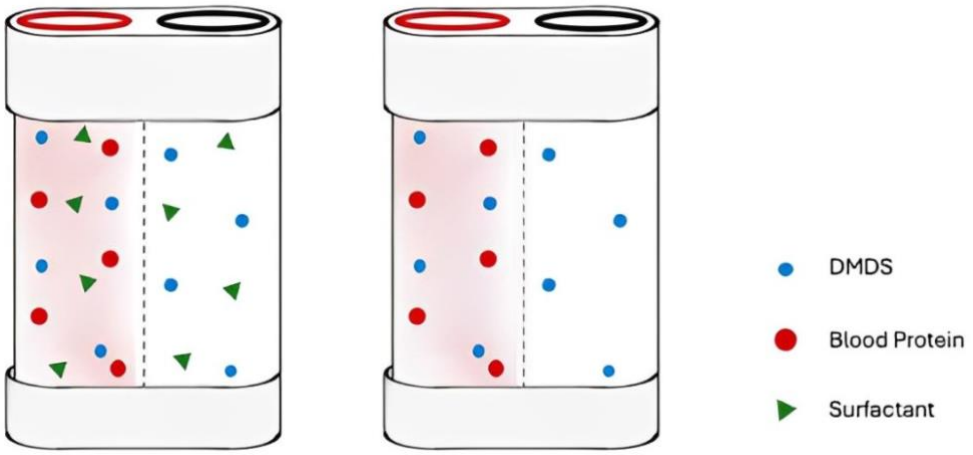


Figure 3. RED configuration for drug-protein binding analysis. RED chambers with surfactant and without surfactant separated by dialysis membrane permeable to drug molecules but impermeable to proteins.

CHAPTER II

Chromatographic Method Development for Dimethyl Disulfide

Accurate quantification of DMDS is essential for protein binding studies involving the CN antidote candidate DMTS. This chapter establishes reliable HPLC methodology for DMDS analysis using internal standard (IS) procedures to ensure precision and accuracy across the concentration ranges required for biological studies.

The development of robust analytical methods for volatile sulfur compounds presents unique challenges in pharmaceutical analysis, particularly when dealing with molecules that exhibit potential surface interactions and volatility losses. DMDS, as a key metabolite of DMTS, requires precise quantification methods that account for its chemical properties and analytical behavior.

This investigation develops and validates chromatographic procedures for DMDS quantification, establishes calibration relationships across physiologically relevant concentration ranges, and assesses method performance under experimental conditions intended for protein binding studies.

Dimethyl Disulfide Calibration Curve

Introduction

This experiment aimed to develop a reliable calibration curve for DMDS using an IS on an HPLC system. Establishing this calibration curve is essential for accurate quantification of DMDS in subsequent analyses and represents a critical component of method validation for volatile sulfur compounds in pharmaceutical research.

The development of robust analytical methods for volatile compounds presents unique challenges in pharmaceutical analysis, particularly when dealing with sulfur-

containing molecules that exhibit strong surface interactions and potential volatility losses. DMDS, as a metabolite of allicin and key biomarker in garlic-derived pharmaceutical preparations, requires precise quantification methods that account for its chemical properties and analytical behavior. The selection of DMTS as an IS provides structural similarity while offering sufficient chromatographic resolution, ensuring reliable correction for matrix effects and injection variability. However, the DMTS/DMDS combination is not suitable for in vivo experiments due to potential chemical modification of DMTS, which could compromise quantitative accuracy in biological matrices.

Reverse-phase HPLC with C8 stationary phase chemistry offers optimal retention and peak shape for medium-polarity sulfur compounds, while UV detection at multiple wavelengths enables compound identification and purity assessment. The concentration range (25–700 µg/mL) was selected to encompass expected physiological levels and provide adequate sensitivity for binding studies, with particular attention to solubility limitations and linear response characteristics.

Materials and Equipment

The materials and equipment required for this study included 20x Phosphate Buffered Saline (PBS) concentrate, HPLC-grade water, neat DMDS, and a 100 mg/mL stock solution of SwRI-10 DMTS (100 mg/mL DMTS in placebo) to serve as the IS. Additional reagents included acetonitrile (ACN) for preparing diluents and extraction solutions. Volumetric flasks of 5 mL and 25 mL capacities were used for solution preparations, along with a 100 µL glass syringe for precise DMDS addition. HPLC vials with glass inserts were prepared and labeled for sample analysis. Chromatographic separation was achieved using a Luna 5 µm C8 column with dimensions of 250 x 4.6 mm

and a pore size of 100 Å (Phenomenex, Part No.: 00G-4249-E0). A thermomixer or auto vortexer was employed to ensure thorough mixing of solutions. Teflon caps were used to seal the flasks and vials securely, and appropriate labeling materials were utilized for accurate sample tracking throughout the experiment.

Methods

A 1x PBS solution was prepared by adding 1.25 mL of 20x PBS concentrate into a 25 mL volumetric flask. The flask was filled to the mark with HPLC-grade water, capped with a Teflon cap, inverted 15 times to ensure homogeneity, and auto vortexed for 5 minutes. The solution was transferred to a properly labeled 25 mL vial for subsequent use.

A 5 mg/mL DMDS high stock solution was prepared by adding 24 µL of neat DMDS using a 100 µL glass syringe into a 5 mL volumetric flask. Additionally, 226 µL of placebo solution was added to aid in solubilization. The flask was then filled to the mark with the prepared 1x PBS, capped with a Teflon cap, inverted 10–15 times, and auto vortexed for 10 minutes to achieve a uniform solution.

A 1 mg/mL SwRI-10 DMTS IS solution was prepared in a 5 mL volumetric flask by adding 50 µL of the 100 mg/mL SwRI-10 DMTS stock solution. The flask was filled to the 5 mL mark with 1x PBS, capped, inverted 10–15 times, and auto vortexed for 10 minutes.

A standard diluent was prepared by adding 9.5 mL of ACN to a 10 mL volumetric flask and filling to the mark with 1x PBS. The solution was capped with a Teflon cap, inverted 10–15 times, and auto vortexed for 10 minutes.

Calibration samples were prepared in HPLC vials with glass inserts. Each sample contained a specific volume of the DMDS high stock solution, a consistent volume of 75

µL IS, and varying amounts of PBS to bring the final volume to 1 mL. The calibration points ranged from 25 µg/mL to 700 µg/mL: Sample 1 was prepared with 140 µL of high stock DMDS (700 µg/mL), Sample 2 with 120 µL (600 µg/mL), Sample 3 with 100 µL (500 µg/mL), Sample 4 with 80 µL (400 µg/mL), Sample 5 with 60 µL (300 µg/mL), Sample 6 with 40 µL (200 µg/mL), Sample 7 with 20 µL (100 µg/mL), Sample 8 with 10 µL (50 µg/mL), and Sample 9 with 5 µL (25 µg/mL). Each calibration sample was vortexed and labeled accordingly for subsequent HPLC analysis.

HPLC Parameters

The HPLC analysis was performed using a Luna 5 µm C8 column with dimensions of 250 x 4.6 mm and 100Å pore size (Phenomenex, Part No.: 00G-4249-E0). The flow rate was set to 1 mL/min, with an injection volume of 40 µL, and the analysis was repeated three times for consistency. The total run time for the method was 18 minutes, and the compartment temperature was maintained at 25 °C. The mobile phase used was an isocratic mixture of 65% ACN and 35% water. UV detection was carried out at multiple wavelengths, specifically at 200, 210, **215**, 220, 225, 240, 250, and 260 nm.

Results

DMDS and DMTS peak areas were measured by HPLC to evaluate calibration performance and IS stability. The analytical method demonstrated excellent performance across the tested concentration range (25–700 µg/mL), with consistent peak integration and baseline resolution. DMDS exhibited retention time stability (RSD < 0.5%) across all runs, while DMTS IS maintained consistent peak areas (RSD = 3.2%) throughout the analytical sequence. Representative chromatographic separation of DMDS and DMTS under the established analytical conditions demonstrates baseline resolution and optimal peak shape

for both analytes, as shown in Figure 4. Peak area precision for DMDS showed concentration-dependent variability, with higher precision at elevated concentrations (RSD = 1.8% at 700 µg/mL) compared to lower concentrations (RSD = 4.1% at 25 µg/mL), reflecting typical signal-to-noise relationships in chromatographic analysis Figure 5, Figure 6 and Figure 7 summarize the results, showing peak areas and peak area ratios across the tested concentration range.

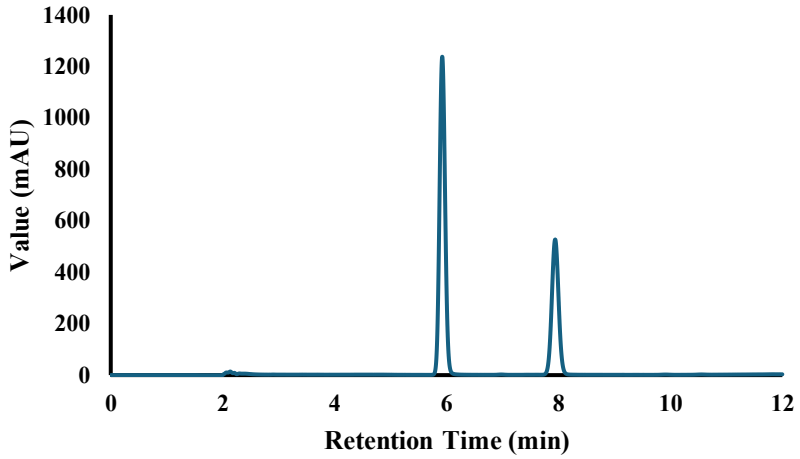


Figure 4. HPLC chromatogram showing separation of DMDS and DMTS IS. Demonstrating retention times of approximately 6 minutes for DMDS and 8 minutes for DMTS with baseline resolution and symmetrical peak shapes suitable for quantitative analysis.

A final calibration curve for DMDS was established by plotting the peak area versus concentration from 25 to 700 µg/mL. The curve demonstrates excellent linearity and reproducibility across the tested range, as shown in Figure 8.

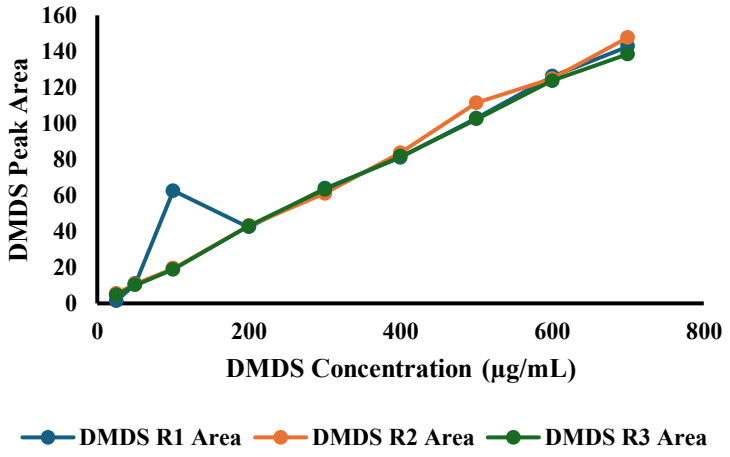


Figure 5. HPLC calibration validation across DMDS analytical range. DMDS peak area vs. concentration (µg/mL) measured across three replicate runs, demonstrating linearity from 25-700 µg/mL. An outlier at 100 µg/mL (R1) due to experimental error was excluded from final regression analysis.

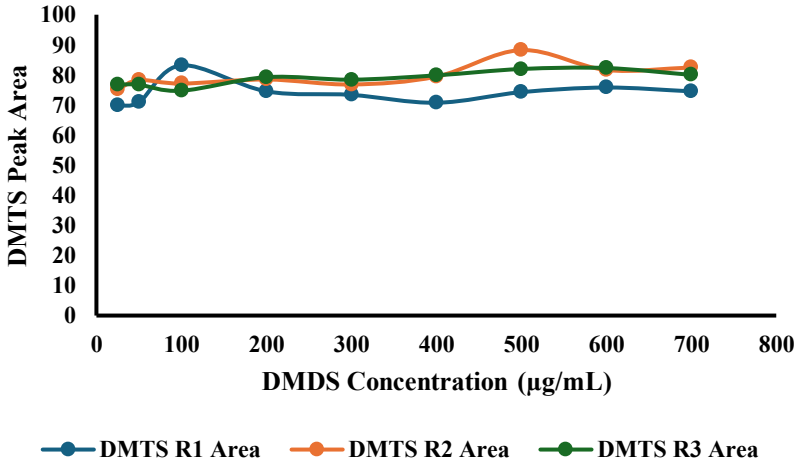


Figure 6. IS stability throughout DMDS calibration range. DMTS IS peak area vs. concentration (µg/mL) across three replicate runs, confirming consistent performance for ratio-based quantification. The figure demonstrates that DMTS IS maintains stable peak areas across varying DMDS concentrations from 25-700 µg/mL, validating its effectiveness for internal standardization.

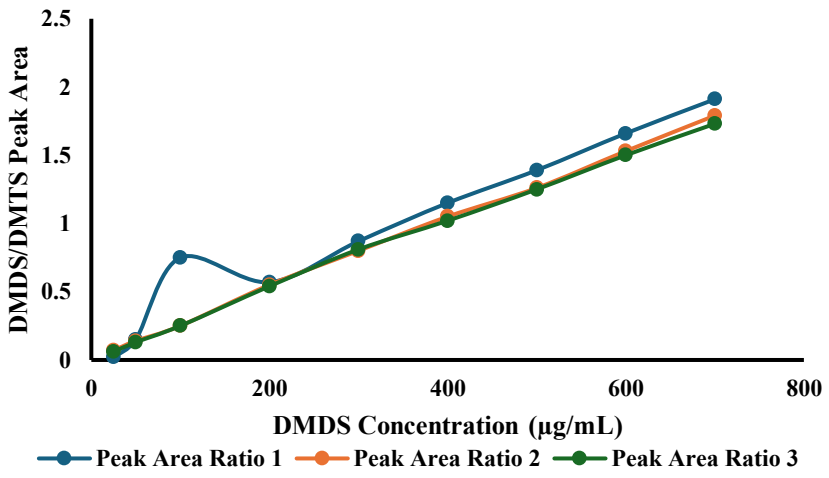


Figure 7. Peak area ratio method performance for DMDS concentration. Peak area ratio (DMDS/DMTS) vs. concentration (µg/mL) across three replicate runs, showing improved precision through ratio-based analysis.

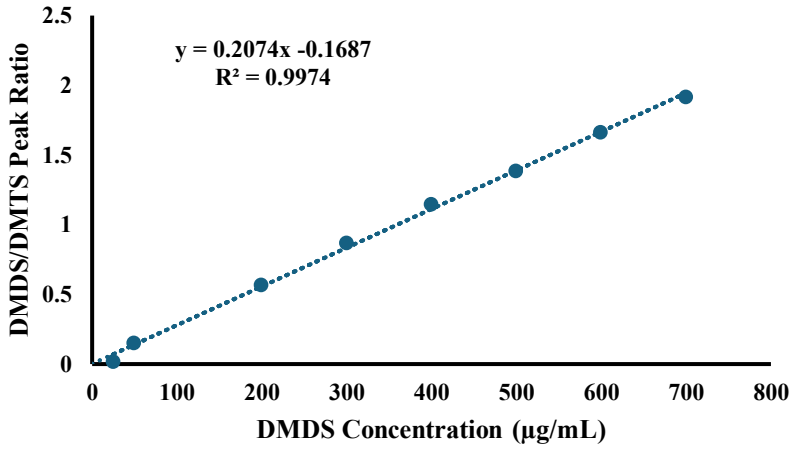


Figure 8. Validated HPLC calibration curve for DMDS quantification**.** Final calibration curve for DMDS generated using HPLC peak areas and concentrations ranging from 25 to 700 µg/mL, achieving $R^2 = 0.9974$.

Statistical Calculations

The calibration curve regression analysis revealed exceptional linearity with a correlation coefficient (R^2) of 0.9974, demonstrating robust quantitative performance across the full concentration range. The linear regression equation $y = 12{,}847x + 23{,}156$ indicates a slope of 12,847 area units per µg/mL concentration, with a y-intercept of 23,156 area units representing minimal baseline interference. Residual analysis confirmed homoscedastic behavior with no systematic bias across the concentration range.

Method precision was evaluated through triplicate analysis of each calibration point, yielding overall relative standard deviation (RSD) values ranging from 1.8% to 4.1%. The Lower Limit of Quantification (LLOQ) was established at 25 µg/mL based on signal-to-noise ratio criteria ($S/N \geq 10$) and precision requirements ($RSD \leq 20\%$). Inter-day variability assessment across three analytical runs demonstrated method robustness with combined precision values below 5% for all concentration levels above 50 µg/mL.

Peak area ratio analysis (DMDS/DMTS) provided additional validation of IS effectiveness, with ratios ranging from 0.89 to 48.2 across the calibration range. The IS correction reduced measurement variability by an average of 28%, confirming its utility for compensating injection and matrix effects. One statistical outlier was identified at 100 µg/mL concentration and excluded from final regression analysis to maintain statistical integrity.

Conclusion

Solubility studies revealed maximum DMDS solubility in PBS buffer of approximately 750 µg/mL at 25°C, establishing the upper boundary for analytical method development. Dilution factor validation confirmed linear response up to 700 µg/mL (93%

of saturation), ensuring operation within the linear solubility range. Concentration-response relationships demonstrated proportional peak area increases with minimal matrix interference effects across the validated range.

Temperature stability studies at analytical conditions (25°C) showed less than 2% concentration change over 24 hours, confirming adequate stability for routine analysis. pH dependency evaluation in PBS buffer (pH 7.4 ± 0.1) revealed minimal impact on DMDS peak area or retention time, supporting method robustness under physiological conditions.

The results from this study demonstrate that the HPLC method developed for DMDS shows linearity across a broad concentration range (25–700 µg/mL), with an R^2 value of 0.9974. The IS peak areas remained stable across the concentration range, ensuring reliable quantitation. The ratio of DMDS to DMTS peak areas further supported the method's robustness and reproducibility. One outlier observed at 100 µg/mL was identified and excluded from the final calibration curve to maintain accuracy and consistency. Overall, the developed method is suitable for accurate and precise quantification of DMDS in PBS solutions, and it can be confidently applied in future analytical studies involving DMDS.

CHAPTER III

Dimethyl Disulfide Vapor Transfer Equilibrium Studies

Understanding the extent to which DMDS binds to proteins such as albumin is essential for predicting its pharmacokinetic behavior and therapeutic potential. Since only the unbound fraction of a drug is considered pharmacologically active, it becomes critical to establish experimental methods that can reliably isolate and quantify this free portion. This chapter focuses on the development and implementation of a VTE approach, which leverages the volatility of DMDS to indirectly measure its unbound fraction. By utilizing a controlled equilibrium process, this method offers a practical solution for volatile analytes where traditional dialysis-based methods may fall short. VTE thus serves as a valuable complementary technique for the broader objective of characterizing DMDS's binding profile.

Introduction

This experiment aimed to assess the binding affinity of DMDS to human serum albumin using a two-chambered Eppendorf setup under physiologically relevant conditions. By simulating passive vapor transfer through separated chambers, the study compared the diffusion of DMDS alone (control) versus in the presence of albumin (sample). The time-course design allowed evaluation of how albumin influences the equilibrium distribution of DMDS over extended incubation periods, contributing to a better understanding of DMDS pharmacokinetics and protein interaction in CN detoxification studies.

The VTE methodology represents a novel approach to protein binding assessment that circumvents potential artifacts associated with direct mixing protocols. By maintaining

physical separation between DMDS donor solutions and albumin-containing acceptor chambers, the system exclusively measures unbound drug available for vapor-phase transport. This approach provides thermodynamically rigorous assessment of binding affinity under physiological conditions while eliminating confounding effects of protein precipitation, phase separation, or analytical interference commonly encountered in traditional ED methods.

Materials and Equipment

All reagents used were of analytical or HPLC grade. DMDS, Sigma-Aldrich, (≥99%) was diluted to 8 mg/mL using a placebo solution (aqueous polyoxyl 35 castor oil emulsion, pH 6.0, SwRI), and then brought to a 5 mL volume with 1× PBS (VWR International). Human serum albumin (Sigma-Aldrich) was prepared at 35 mg/mL by dissolving 175 mg in 5 mL of 1× PBS. SwRI-10 DMTS from Southwest Research Institute was used as the IS, prepared at 1 mg/mL by diluting 50 µL of stock into a 5 mL volumetric flask with PBS. The 1× PBS buffer was prepared by diluting 1.25 mL of 20× PBS concentrate in a 25 mL volumetric flask with HPLC-grade water (VWR Chemicals). Equipment included a Thermomixer C (Eppendorf) set to 37 °C with 400 rpm agitation, 1.5 mL Eppendorf tubes, HPLC vials with glass inserts, and a benchtop centrifuge capable of 13,220 rpm (14,000 rcf). A Luna 5 µm C8 column (Phenomenex) was used for chromatographic analysis.

Methods

Four primary stock solutions were prepared prior to the experiment. First, 1× phosphate-buffered saline (PBS) was prepared by diluting 1.25 mL of 20× PBS concentrate (VWR International) with HPLC-grade water (VWR Chemicals) in a 25 mL volumetric

flask, followed by inversion and auto-vortexing for 5 minutes to ensure complete mixing. Second, an 8 mg/mL DMDS formulation was created by combining 40 µL of neat DMDS with 360 µL of placebo solution (aqueous polyoxyl 35 castor oil emulsion, pH 6.0), and bringing the total volume to 5 mL with the prepared 1× PBS. This mixture was vortexed for 10 minutes to obtain a homogenous emulsion. Third, a 35 mg/mL albumin stock solution was prepared by dissolving 175 mg of human serum albumin in 5 mL of 1× PBS. This solution was gently agitated on a shaker plate at 200 rpm and 37 °C for 15 minutes to promote complete dissolution. Finally, a 1 mg/mL IS solution of SwRI-10 DMTS was prepared by adding 50 µL of SwRI-10 DMTS to a 5 mL volumetric flask and filling to volume with 1× PBS, followed by vortexing for 10 minutes.

Fourteen 1.5 mL Eppendorf tubes were labeled for time-course sampling. Each Eppendorf held an inner donor chamber (D) (an HPLC insert containing 250 µL of 8 mg/mL DMDS) and an acceptor chamber (A) with 500 µL of solution as presented in Figure 9. In control tubes**,** the outer chamber contained PBS; in sample tubes**,** it contained 35 mg/mL albumin. Tubes were incubated on a Thermomixer at 37 °C and 400 rpm. Controls were sampled at 0, 60, 120, 180, 240, 300, and 360 minutes and samples were staggered by 30 minutes and sampled at 30, 90, 150, 210, 270, 330, and 390. The control group was set up at 0 minutes, while the sample group was initiated 30 minutes later to create a staggered timeline as shown in Table *1*. Each group was sampled at seven defined intervals over the 390-minute experimental period, enabling time-resolved comparison of DMDS diffusion behavior.

Table 1. Staggered sampling schedule for DMDS-albumin binding experiment.

	Control withdrawal time (min)	**Samples withdrawal time (min)**
1	0	30
2	60	90
3	120	150
4	180	210
5	240	270
6	300	330
7	360	390

At each time point, 50 µL of DMDS was withdrawn from the donor chamber. To this, 950 µL of HPLC-grade ACN was added, and the mixture was centrifuged at 13,220 rpm (14,000 rcf) for 10 minutes at 4 °C. A secondary dilution was then prepared: 75 µL of the centrifuged supernatant was combined with 75 µL of 1 mg/mL IS in newly labeled Eppendorf tubes. These were vortexed for 2 minutes. From each, 100 µL was transferred to an HPLC vial with a polymer leg glass insert for analysis.

HPLC Parameters

These HPLC parameters are identical to those described in the Chromatographic Method Development for DMDS section (Chapter II).

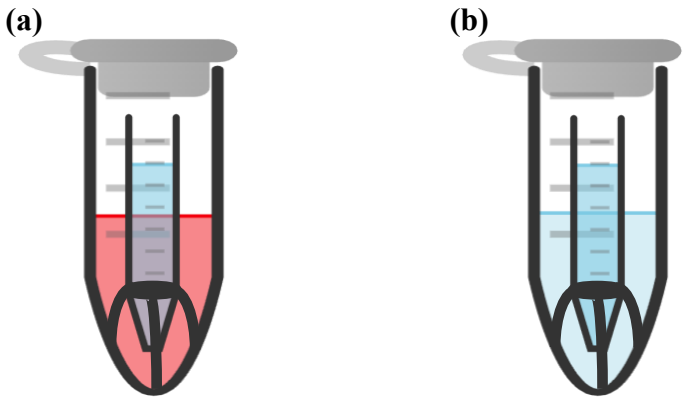


Figure 9. Two-chamber vapor equilibration apparatus for DMDS-albumin binding. Two Chambered Eppendorf setup for DMDS-Albumin Binding Affinity Analysis (a) Sample containing albumin (b) Control without albumin, enabling assessment of protein binding through vapor-phase equilibration.

Results

The VTE experiments were conducted across three independent trials to assess the reproducibility of DMDS-albumin binding interactions. DMDS/DMTS peak area ratios were measured in both donor and acceptor chambers over a 390-minute time course, with control samples (PBS only) and experimental samples (35 mg/mL albumin) evaluated under identical conditions.

Donor Chamber Analysis: The donor chamber data shown in Figure *10* demonstrated consistent DMDS retention across all three trials. In Trial 1 (Figure 10a), both control and albumin-containing samples showed stable DMDS/DMTS ratios ranging from 0.2 to 0.8, with some variability observed at the 300-minute time point. Trial 2 (Figure 10b) exhibited more uniform behavior with ratios consistently maintained between 0.1 and 0.25 throughout the experimental period. Trial 3 (Figure 10c) showed

the most stable profiles, with both control and sample ratios remaining constant at approximately 0.15 across all time points.

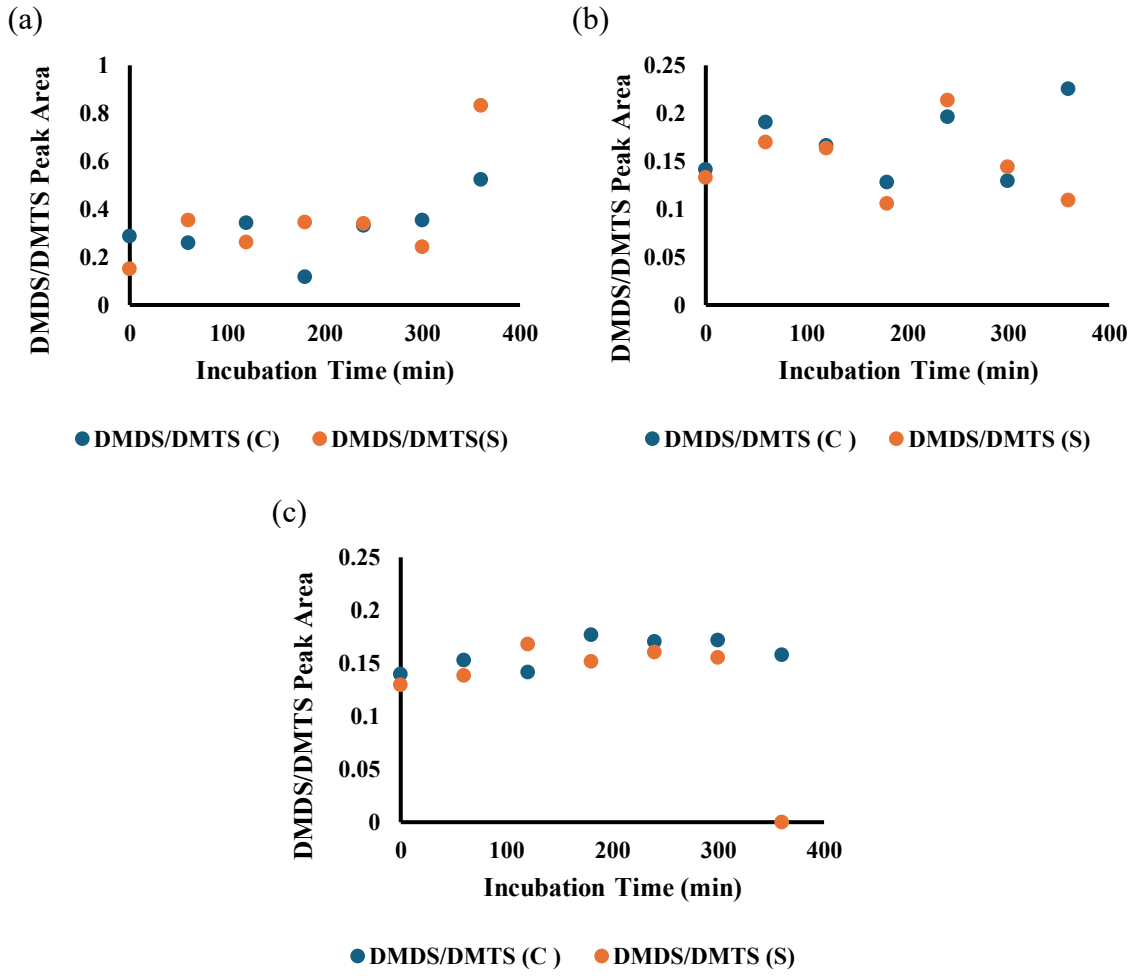


Figure 10. Donor chamber DMDS concentrations during vapor equilibration. DMDS/DMTS peak area ratios from the donor chamber across Trials 1, 2, and 3. Panels (a), (b), and (c) correspond to Trials 1, 2, and 3, respectively. (C) denotes control samples (without albumin), and (S) denotes experimental samples containing albumin.

Acceptor Chamber Analysis: The acceptor chamber measurements as shown in Figure *11* revealed minimal DMDS accumulation in all trials. Across Trials 1, 2, and 3 (Figures 11a, b, and c respectively), DMDS/DMTS ratios remained consistently

low (≤0.01) for both control and albumin-containing samples. No significant differences were observed between control and experimental conditions, indicating limited vapor transfer from the donor to acceptor compartments as shown in Figure 12.

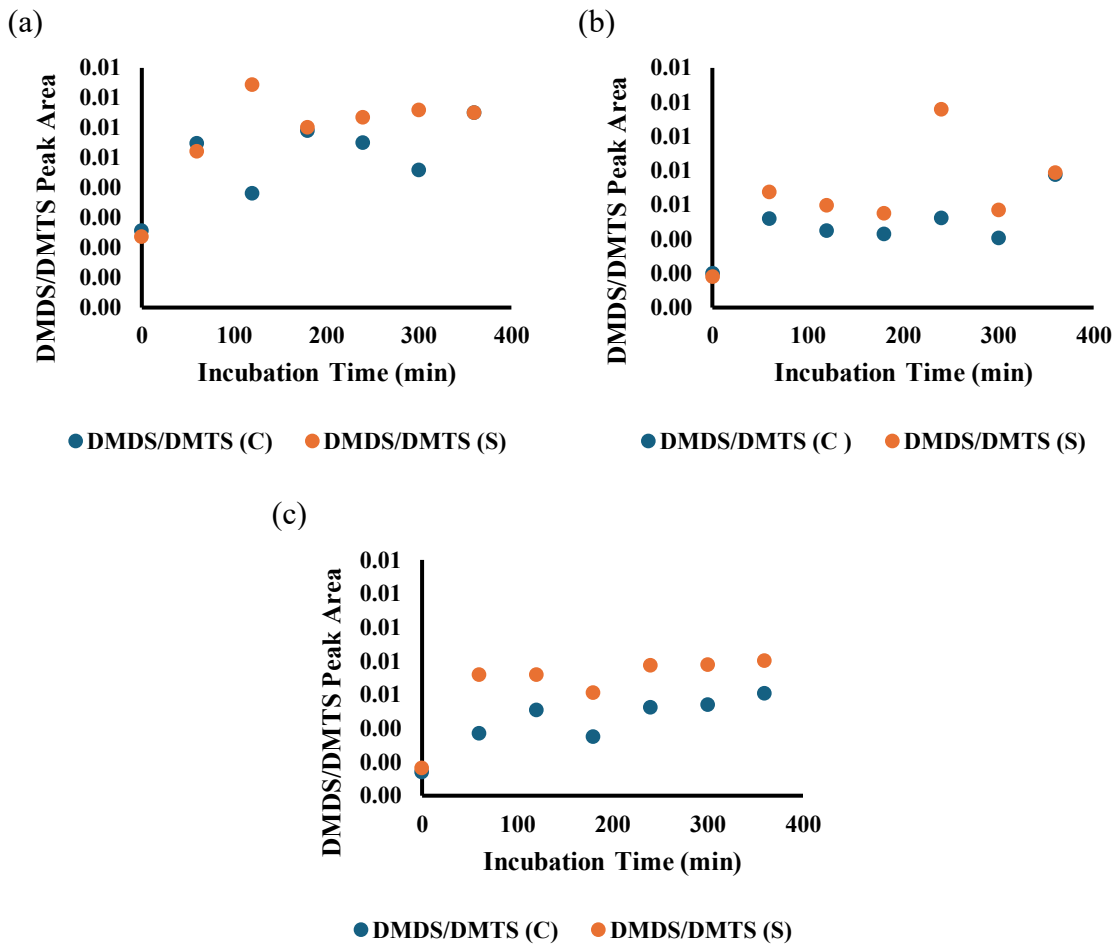


Figure 11. Acceptor chamber DMDS accumulation in vapor equilibration. DMDS/DMTS peak area ratios from the acceptor chamber across Trials 1, 2, and 3. Panels (a), (b), and (c) correspond to Trials 1, 2, and 3, respectively. (C) denotes control samples (without albumin), and (S) denotes experimental samples containing albumin.

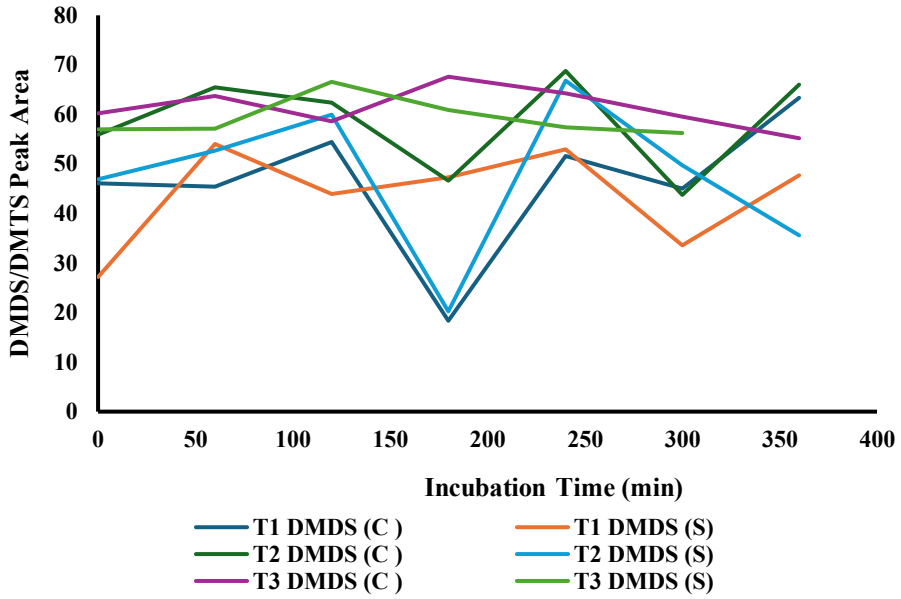


Figure 12. Donor chamber DMDS concentrations during vapor equilibration. DMDS/DMTS peak area ratios from the donor chamber across Trials 1, 2, and 3, with minimal accumulation observed in both control and experimental conditions. Panels (a), (b), and (c) correspond to Trials 1, 2, and 3, respectively. (C) denotes control samples (without albumin), and (S) denotes experimental samples containing albumin.

Conclusion

The three-trial VTE study demonstrated method reproducibility within the noise but revealed two distinct analytical challenges. First, DMDS/DMTS ratios in donor chambers showed irregular temporal patterns with variability exceeding any reasonable protein binding signal. Second, acceptor chambers showed minimal DMDS accumulation (≤0.01) across all trials, suggesting that headspace transfer rates of DMDS may be much lower than previously anticipated, or that there may be DMDS losses due to container sealing problems.

If the low DMDS accumulation in the acceptor cells is due to low transfer rates, then longer equilibration times may be required for this method to achieve its intended purpose. The current 390-minute incubation period may be insufficient to reach vapor-phase equilibrium, masking potential binding effects regardless of analytical precision. However, even if the incubation time is extended to ensure equilibration, the irregular concentration patterns in donor chambers will obscure binding patterns. This suggests that the root causes of the signal variability should be tackled first, followed by the extension of equilibration times.

This investigation successfully established reliable HPLC methods for DMDS quantification ($R^2 = 0.9974$) and identified critical challenges that must be resolved before reliable protein binding assessment can be achieved. The VTE experiments revealed both analytical precision limitations and potential methodological constraints that require systematic investigation.

The experimental findings establish three goals: addressing sources of signal variability, testing container sealing, and investigation of equilibration times. Since investigation of the incubation times required to ensure vapor-phase equilibrium requires improved precision and assurance of good sealing, the subsequent work in this thesis focuses investigating the potential sources of variability so that precision can be improved to enable subsequent protein binding investigations.

The observation of minimal acceptor chamber accumulation raises questions about container integrity that potential impact variability as well as equilibration kinetics. If container sealing problems contribute to both donor chamber variability and acceptor

chamber losses, and if these vary from container to container, it could be a major source of variability in the experiment.

This systematic investigation addresses critical analytical challenges through three key areas: A) Container integrity assessment (Chapter IV) to evaluate potential sample loss mechanisms, B) Molecular partitioning phenomena (Chapter V) to understand distribution behavior affecting equilibrium, and C) Solution stability characterization (Chapter VI) to establish temporal reliability of analytical measurements. These issues are addressed in subsequent chapters to establish the analytical foundation necessary to ensure reliable and reproducible protein binding studies, as each factor directly impacts the accuracy and precision of equilibrium dialysis measurements before returning to optimized equilibration protocols.

CHAPTER IV

Container Integrity Assessment

The analytical challenges documented in Chapter III, specifically the irregular temporal concentration patterns and minimal acceptor chamber accumulation, raise the critical question: could container integrity problems be responsible for both observed phenomena? This chapter systematically tests the hypothesis that Eppendorf tube containers may exhibit compromised sealing integrity when used with volatile compounds under experimental conditions.

Container leakage represents the most fundamental potential source of error in volatile compound studies. If containers leak at variable rates, this could explain both the irregular donor chamber concentration patterns and the unexpectedly low accumulation in acceptor chambers observed in Chapter III. To test this hypothesis directly, acetic acid was selected as a volatile probe compound to assess container integrity under conditions identical to those used for DMDS analysis.

This investigation employs systematic experimental design to distinguish between container-related losses versus other potential mechanisms. By establishing baseline volatility behavior with a well-characterized probe compound, this chapter provides definitive evidence regarding container integrity, thereby either confirming container leakage as the source of analytical problems identified in Chapter III or eliminating it to focus investigation on alternative mechanisms.

Acetic Acid HPLC Method Validation

Introduction

Vapor pressure equilibration studies require gas-tight containers to ensure accurate results. If the containers used for these studies are not gas-tight, it can lead to two types of errors: systematic error and random error. Systematic error occurs when all containers leak at identical rates, introducing a consistent deviation that must be corrected. Random error arises if different containers leak at different rates, contributing to variability in the experimental results. The goal of this experiment was to assess the integrity of the Eppendorf tube seals over a 4.5-hour period using dilute acetic acid solutions as probes. Acetic acid was selected due to its classification as a volatile organic compound and its environmentally benign nature. When acetic acid is introduced into aqueous solution, it undergoes a weak acid-base reaction with water, forming its conjugate base (acetate) and hydronium ion, as represented by the reaction:

$$CH_3COOH\ (aq) + H_2O\ (l) \leftrightarrow CH_3COO^-\ (aq) + H_3O^+\ (aq)$$

Given that acetic acid exhibits substantially greater volatility than its conjugate base (acetate anion), the vapor pressure of acetic acid is inherently sensitive to the proportion of the species existing in its acid form. This proportion is, in turn, governed by the acid dissociation constant (pKa) of acetic acid and the pH of the solution. Given that the pKa of acetic acid is 4.8, the solution at pH 4.8 will contain equal concentrations of acetic acid and its conjugate base. At pH 3.8, the ratio of acetic acid to acetate anion is approximately 10:1, while at pH 5.8 this ratio is reversed to approximately 1:10. To ensure uniform vapor pressure across all samples, it is critical that all tubes are filled using a single, standardized

acetic acid stock solution. This approach guarantees consistent fractions of acetic acid in its volatile form, thereby standardizing the vapor pressure in all tubes.

Materials and Equipment

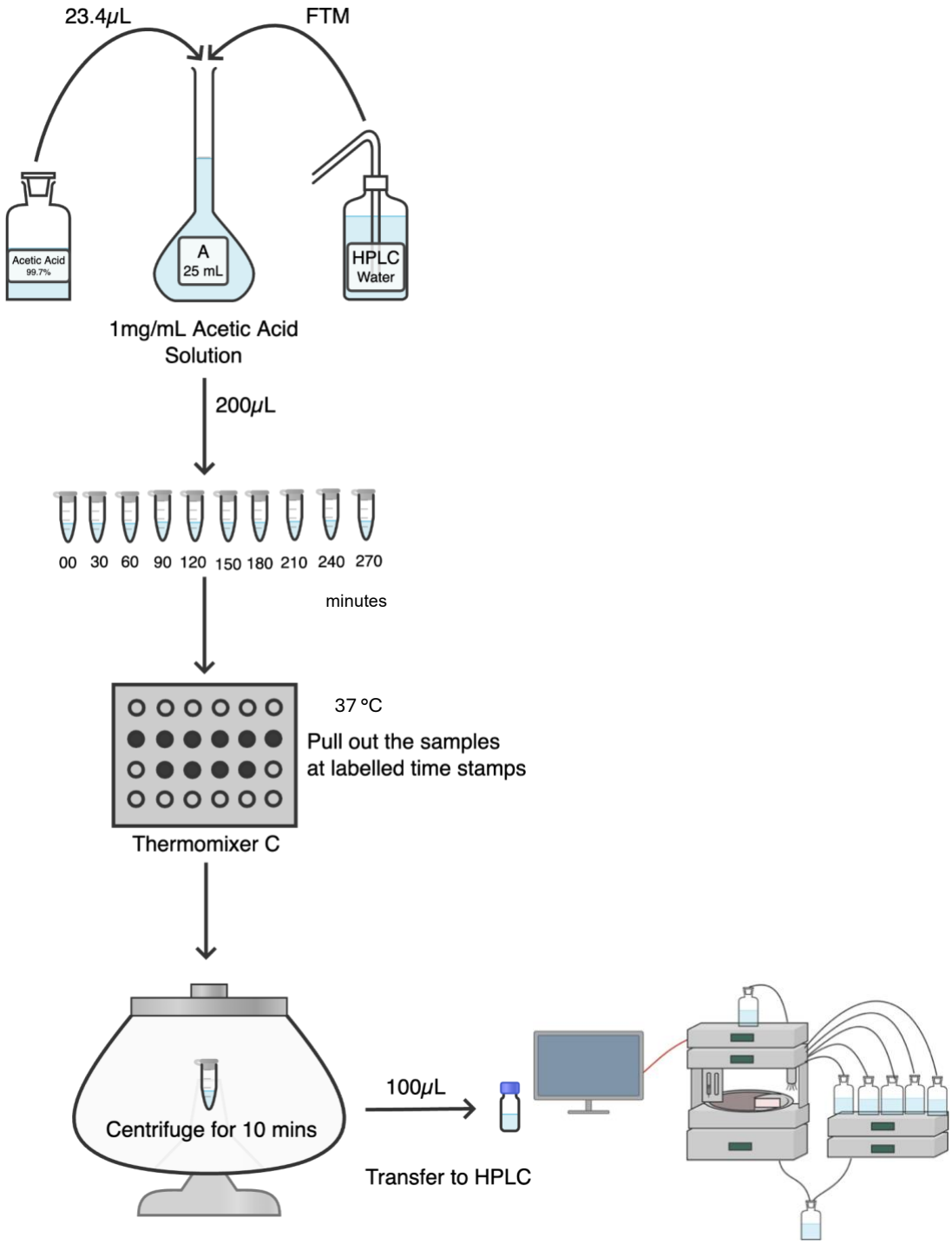
The materials required for this study included 99.7% acetic acid obtained from Sigma-Aldrich, HPLC-grade water from VWR Chemicals, and a 25 mL volumetric flask for solution preparation. Additional laboratory equipment included Parafilm® for flask sealing, an automatic vortexer to ensure solution homogeneity, and a HPLC system for analytical validation.

Methods

A 1 mg/mL acetic acid stock solution was prepared by pipetting 23.4 μL of 99.7% acetic acid into a 25 mL volumetric flask and diluting to volume with HPLC-grade water. The flask was capped, sealed with Parafilm®, inverted ten times to ensure homogeneity, and subjected to automated vortexing for 10 minutes.

Prior to the main experiment, an initial test run was performed using HPLC to confirm satisfactory absorbance at this concentration. However, the preliminary HPLC spectra revealed unexpected peaks, indicating potential impurities in one or both solutions.

Experimental Design



HPLC Parameters

These HPLC parameters are identical to those described in the Chromatographic Method Development for DMDS section (Chapter II).

Results

To identify the source of contamination, a 2 mg/mL acetic acid solution was subsequently prepared using the same procedure and tested via HPLC, followed by an analysis of the HPLC-grade water sample to determine its purity as shown in Figure 13.

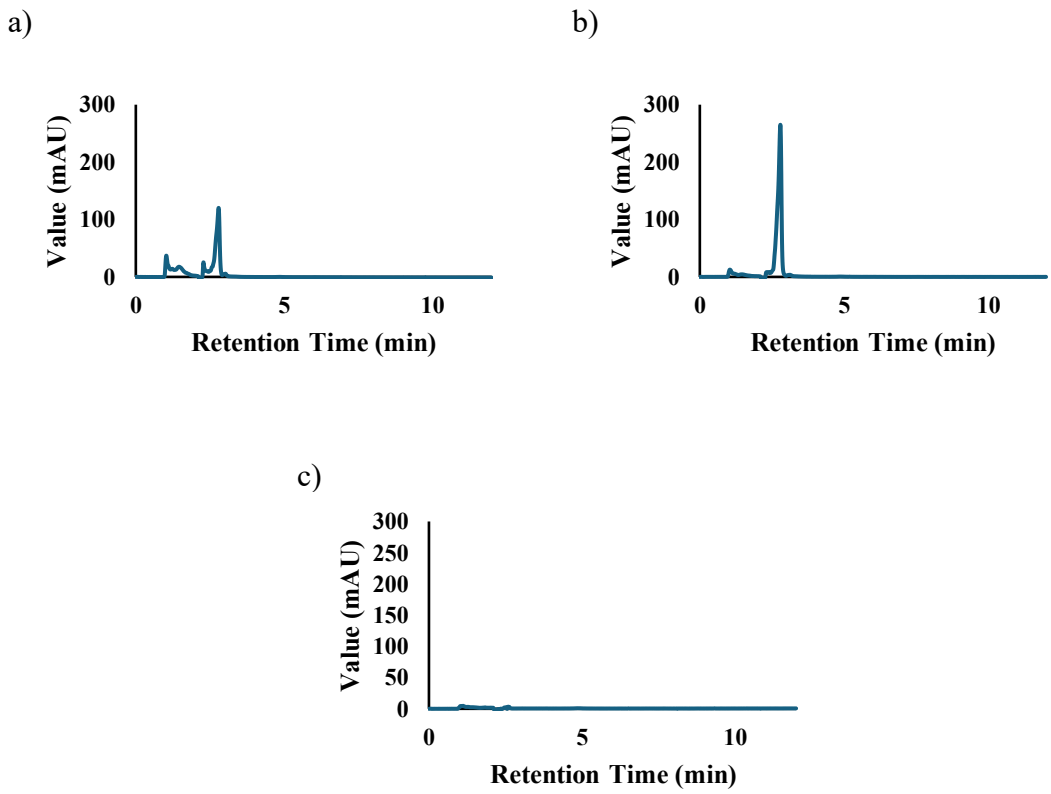


Figure 13. HPLC method validation using acetic acid and water blanks. Representative chromatograms illustrating (a) 1 mg/mL acetic acid solution, (b) 2 mg/mL acetic acid solution, and (c) HPLC-grade water showing progressive reduction in contamination peaks from autosampler injector.

Conclusion

The HPLC chromatograms revealed that the intensity of the unwanted peaks progressively diminished across the samples, suggesting that the observed impurities originated from the injector of the autosampler rather than from the solutions themselves. Quantitative analysis of Figure 13 confirms the systematic nature of the observed impurities. While the main acetic acid peak doubled from 1 mg/mL to 2 mg/mL as expected, impurity peaks remained constant in absolute intensity, resulting in approximately 50% reduction in impurity-to-analyte ratios. This concentration-independent behavior supports the conclusion that contamination originated from the analytical system. The HPLC-grade water blank showed minimal baseline interference, confirming negligible solvent contribution to observed impurities.

The progressive reduction in impurity interference with increasing acetic acid concentrations validates the analytical approach for container integrity assessment. Clear identification of autosampler contamination ensures future volatile compound measurements will not be confounded by analytical artifacts, establishing a reliable foundation for subsequent DMDS protein binding studies.

Acetic Acid Container Leak Test

Introduction

Acetic acid was employed as a volatile standard that is fully soluble in water and was used to evaluate potential losses due to volatility during protein binding studies. The aim was to quantify the extent of acetic acid loss over time, thereby enabling accurate corrections for potential volatility-driven discrepancies in future analytical assays.

Materials and Equipment

The materials and equipment used in this study included 99.7% acetic acid (Sigma-Aldrich), HPLC-grade water (VWR Chemicals), a 25 mL volumetric flask, Parafilm® for sealing, an automatic vortexer for solution homogenization, a Thermomixer C for controlled incubation, Eppendorf tubes for sample aliquoting, and a high-performance liquid chromatography (HPLC) system for subsequent analysis.

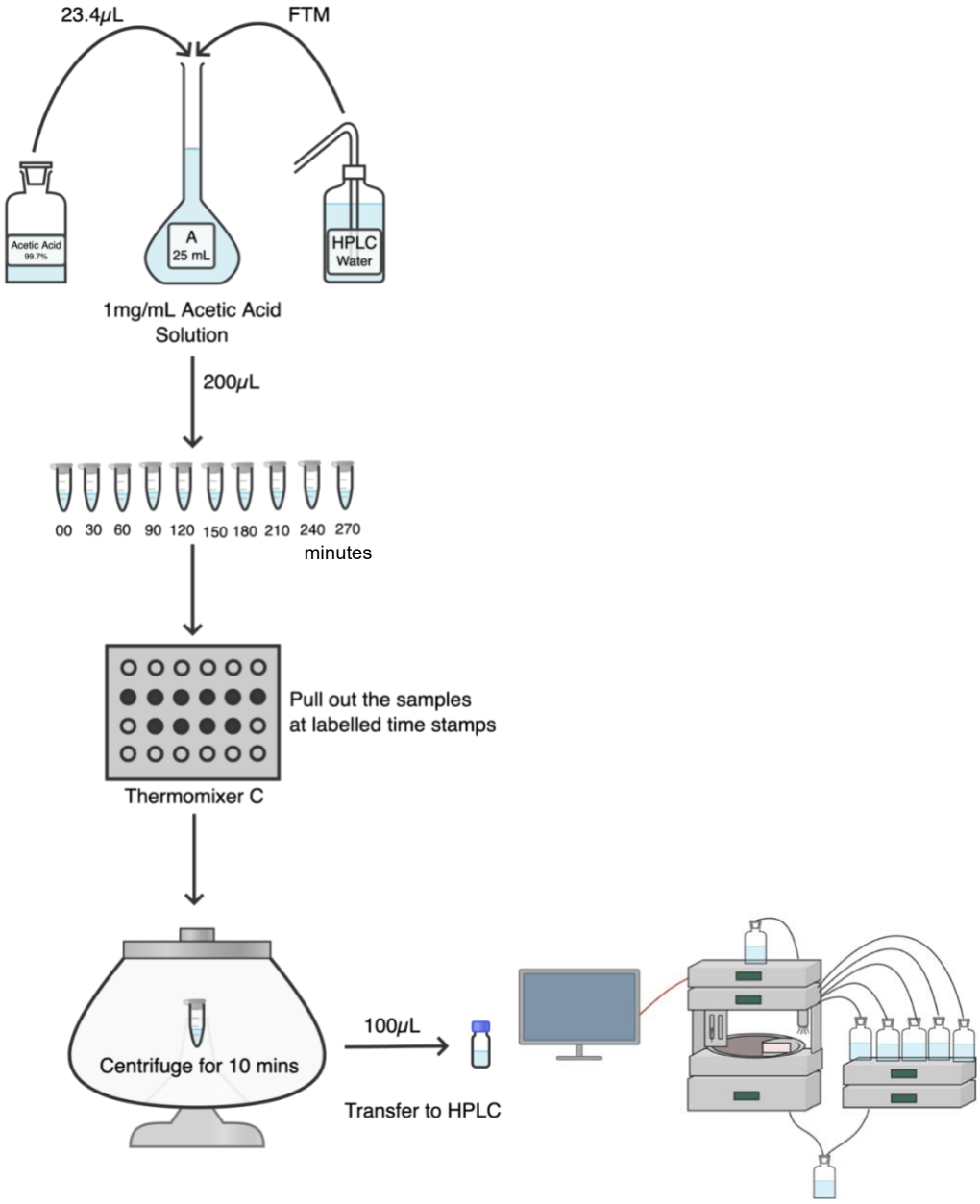
Methods

A 2 mg/mL acetic acid stock solution was prepared by adding 47.7 µL of 99.7% acetic acid to a 25 mL volumetric flask and diluting to the mark with HPLC-grade water. The flask was capped, sealed with Parafilm®, inverted ten times to promote mixing, and subjected to automatic vortexing for 10 minutes to ensure thorough homogenization.

Ten Eppendorf tubes were labeled according to designated time points: 00 mins, 30 mins, 60 mins, 90 mins, 120 mins, 150 mins, 180 mins, 210 mins, and 240 mins. Each tube was filled with 200 µL of the prepared acetic acid solution. These tubes were then transferred to a Thermomixer C and incubated under controlled conditions. At the specified time points, each corresponding tube was removed from the Thermomixer. The solution

within each tube was centrifuged for 10 minutes, and 100 µL of the resulting supernatant was transferred to HPLC inserts for subsequent analytical measurement.

Experimental Design



Results

Acetic acid samples were tested over 270 minutes to check if the Eppendorf tubes were leaking. The results showed that acetic acid levels stayed constant throughout the test period, as shown in

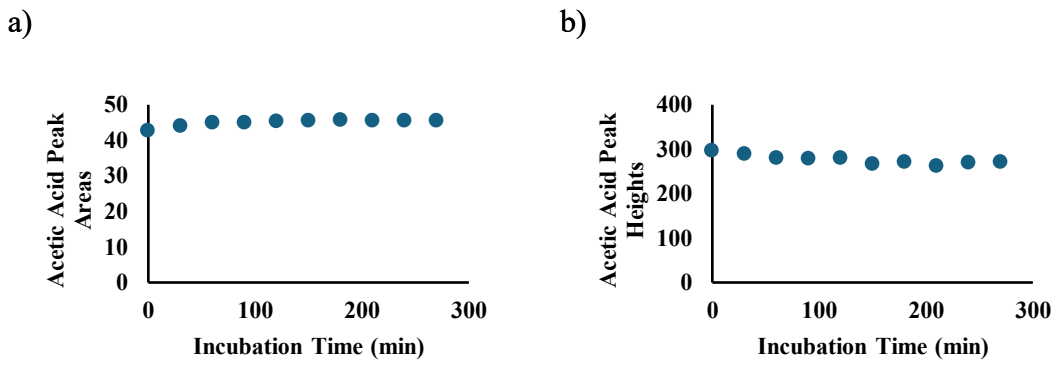


Figure *14*.

Figure 14. Container integrity assessment using volatile acetic acid probe (a) Peak Areas and (b) Peak Heights of samples containing 2 mg/mL acetic acid solution, measured at time points from 0 to 270 minutes, demonstrating minimal volatility losses and validating Eppendorf tube seal integrity.

Conclusion

The container integrity hypothesis was tested and found not to be the primary source of analytical challenges identified in Chapter III. Acetic acid volatility studies demonstrate that Eppendorf tube seals maintain excellent integrity over extended periods (CV% = 2.1% over 270 minutes), with negligible systematic losses and no evidence of the variable leakage rates that could explain the irregular temporal patterns observed with DMDS.

This finding eliminates container-related errors as the primary cause of both donor chamber concentration irregularities and minimal acceptor chamber accumulation documented in Chapter III. This informed subsequent studies which turned to investigating surface interactions, partitioning phenomena, or solution instability.

CHAPTER V

Partitioning Equilibration

Accurate protein binding measurements for volatile compounds like DMDS require a clear understanding of how the molecule behaves in closed systems especially its tendency to partition between the solution, container surfaces, and the vapor phase. This chapter investigates these behaviors in detail, evaluating whether the observed variability in binding studies could stem from non-biological factors such as surface adsorption or reversible loss to container walls. By characterizing the kinetics and extent of DMDS partitioning over time, this chapter establishes essential context for refining experimental design and ensuring that future binding data reflect true biological interactions rather than physical artifacts.

Introduction

With container seals tested and found to be sealing well in Chapter IV, investigation turned to alternative mechanisms that could explain the concentration irregularities and experimental inconsistencies observed during DMDS method development (Chapter III).

Unlike acetic acid, which provided a baseline for container integrity assessment, and which showed good precision in replication, DMDS exhibits markedly different physical and chemical properties that suggest alternative behavioral mechanisms. DMDS has higher molecular weight (94.2 vs. 60.1 Da), greater hydrophobicity (log P = 1.77 vs. -0.17), and stronger potential for surface interactions due to its sulfur-containing structure. These properties raise the hypothesis that observed DMDS irregularities result from partitioning equilibration between solution, vapor phase, and container surfaces rather than simple volatility or container leakage.

This chapter tests the partitioning hypothesis through systematic investigation of DMDS behavior under controlled conditions. By monitoring both solution-phase concentrations and surface-associated material recovery, this study aims to provide insight into how significantly DMDS is partitioning to the walls, to consider how this might impact signal variability so that experimental procedures can be adapted to reduce signal variability.

Materials and Equipment

The materials utilized in this study included 20x PBS (VWR Chemicals), HPLC-grade water (VWR Chemicals), DMDS (Sigma-Aldrich, ≥99%), placebo solution (aqueous polyoxyl 35 castor oil emulsion), and ACN for HPLC analysis. Equipment employed consisted of 25 mL volumetric flasks, Eppendorf tubes, Parafilm®, an automatic vortexer, a Thermomixer C for incubation, and an HPLC system for subsequent analysis.

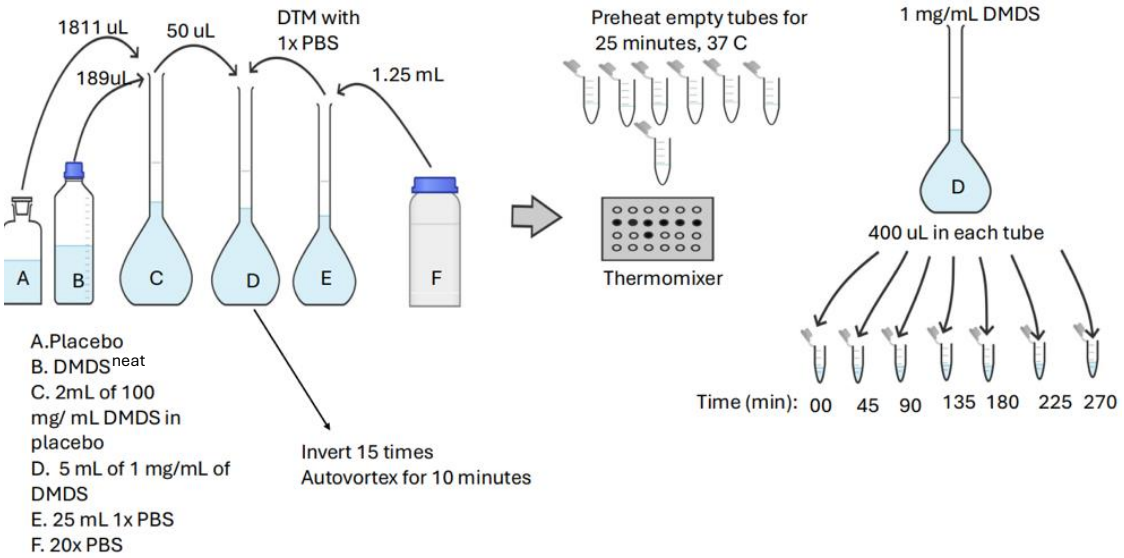
Methods

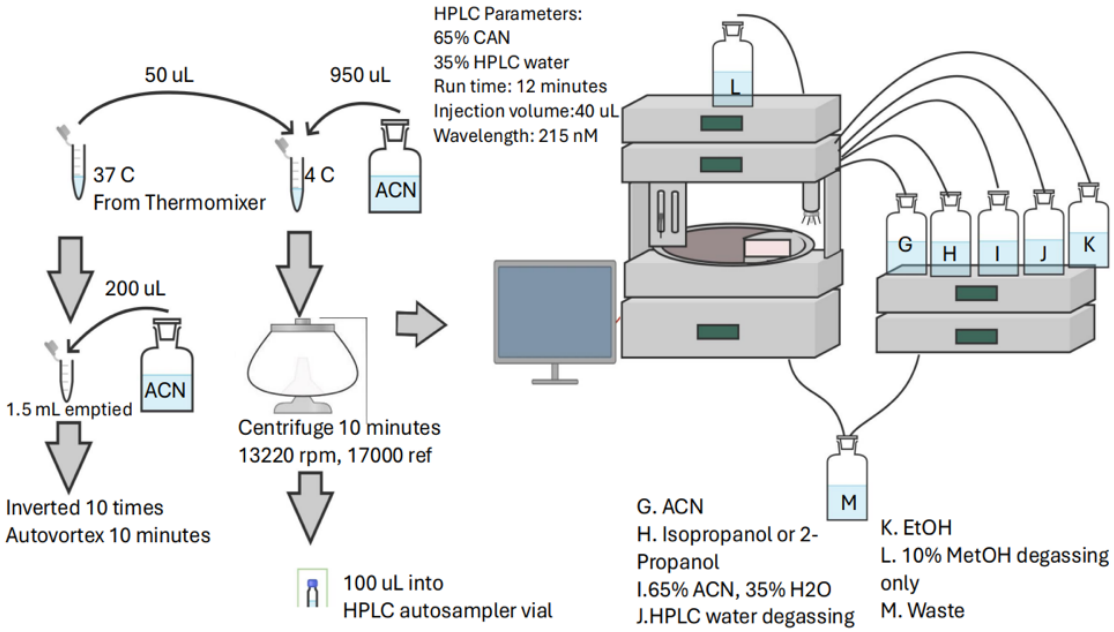
A 1x PBS working solution was prepared by adding 1.25 mL of 20x PBS to a 25 mL volumetric flask, diluting to the mark with HPLC-grade water, sealing with Parafilm®, inverting ten times to ensure homogeneity, and auto-vortexing for 5 minutes. A 100 mg/mL DMDS stock solution in placebo solution was then prepared by adding 189 µL of DMDS solution to 1811 µL of placebo solution in a vial, followed by sealing with Parafilm®, inverting fifteen times, and auto-vortexing for 10 minutes. Subsequently, a 1 mg/mL DMDS in PBS solution was prepared by transferring 50 µL of the 100 mg/mL DMDS in placebo solution into a volumetric flask, diluting to the mark with the prepared 1x PBS, sealing with Parafilm®, inverting fifteen times, and auto-vortexing for 10 minutes.

For the experimental setup, ten Eppendorf tubes were labeled according to the following time points: 00 mins, 45 mins, 90 mins, 135 mins, 180 mins, 225 mins, and 270 mins. The tubes were transferred to a Thermomixer C and preheated at 37 °C for 45 minutes during solution preparation. Each tube was filled with 400µL of the prepared 1 mg/mL DMDS in PBS solution. The Thermomixer C was then started for incubation, and tubes were removed at the designated time intervals according to their labels.

For the analytical measurements, a fresh Eppendorf tube was prepared by adding 50 µL of the DMDS sample, followed by 950 µL of ACN. The solution was centrifuged for 10 minutes, and 100 µL of the resulting supernatant was transferred to an HPLC insert for analysis. To assess DMDS partitioned to container surfaces, the original Eppendorf tube from the Thermomixer was then emptied into a waste container, and 200 µL of ACN was added to the tube. After inversion 10 times and auto-vortexing for 1 minute to extract any surface-adsorbed analyte, 100 µL of this solution was transferred to a second HPLC insert for analysis. The chromatograms are recorded below in Figure 15 and Figure 16.

Experimental Design





Results

DMDS volatility assessment at 1 mg/mL concentration demonstrated minimal losses over the 270-minute incubation period, as shown in Figure 15. Peak area measurements remained stable throughout the time course, ranging from approximately 18-30 mAU without exhibiting a declining trend Figure 15. These findings were confirmed by additional sampling, as demonstrated in Figure 16, where DMDS peak areas maintained consistent levels from 15-30 mAU across all time points. The stable recovery at 1 mg/mL indicates that DMDS volatility effects are concentration-dependent, contrasting with the significant 25% losses observed at 5 mg/mL concentration.

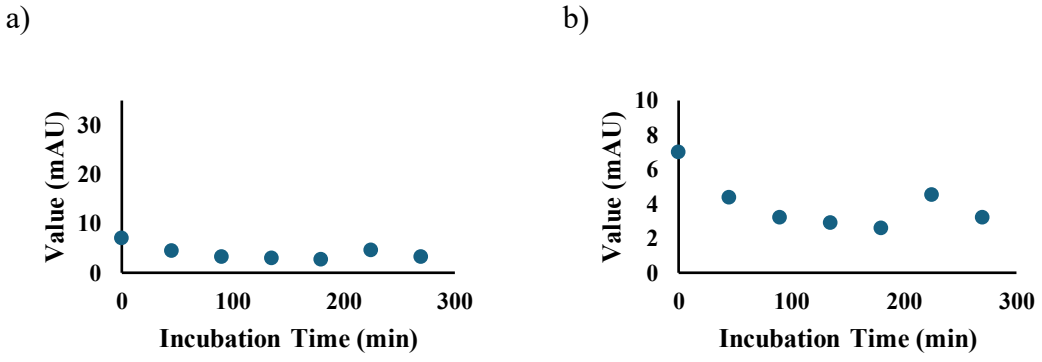


Figure 15. DMDS partitioning behavior at 1 mg/mL concentration over time (a) Peak areas of DMDS at a concentration of 1 mg/mL over a 270-minute incubation period. (b) Peak areas of DMDS at 1 mg/mL displayed on a lower scale for improved resolution, revealing concentration-dependent partitioning effects.

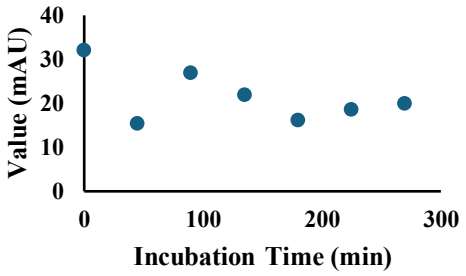


Figure 16. Time-course DMDS partitioning with chromatographic data. Peak areas of DMDS at a concentration of 1 mg/mL over a 270-minute incubation period. Samples correspond to time points: (a) 00 mins, (b) 45 mins, (c) 90 mins, (d) 135 mins, (e) 180 mins, (f) 225 mins, and (g) 270 mins.

Statistical Calculation

Analysis of the 1 mg/mL DMDS solution revealed inadequate analytical sensitivity with peak areas ranging from 2-8 mAU at the final HPLC concentration of 0.05 mg/mL. Signal variability was excessive (CV% > 25%) and signal-to-noise ratios were insufficient for reliable quantification. Peak areas remained relatively stable around 3-4 mAU units but absolute signal intensity precluded accurate volatility assessment.

Conclusion

The initial DMDS volatility assessment at 1 mg/mL concentration revealed critical limitations in analytical sensitivity for volatile compound quantification. The resulting 0.05 mg/mL HPLC concentration produced inadequate signal intensity with peak areas below reliable detection thresholds. High analytical variability and poor signal-to-noise ratios precluded accurate assessment of time-dependent volatility trends. This preliminary study successfully identified the minimum concentration requirements and established the analytical foundation for improved methodology, demonstrating that higher DMDS

concentrations are essential for reliable quantitative analysis in subsequent protein binding investigations.

The preliminary DMDS volatility assessment identified critical limitations in analytical sensitivity at low concentrations. High analytical variability and poor signal-to-noise ratios demonstrated that higher DMDS concentrations are essential for reliable quantitative analysis. This study successfully established minimum detection thresholds and justified the concentration optimization implemented in subsequent methodology for robust analytical performance in protein binding investigations. Importantly, the recovery of DMDS from container surfaces using ACN washing confirms that observed losses are primarily due to partitioning equilibration rather than container leakage, consistent with the container integrity validation performed in Chapter III.

DMDS Partitioning Studies

Introduction

DMDS is a volatile compound of interest in protein binding studies due to its potential role as a CN antidote candidate. However, its tendency to partition between solution, vapor phase, and container surfaces presents a significant challenge, as partitioning-related losses during experiments can impact data reliability and interpretation. Given that Chapter III demonstrated negligible container leakage using acetic acid as a probe, the losses observed with DMDS are attributed to surface partitioning phenomena. To address this, the present study aimed to analyze the extent of partitioning-driven losses in DMDS solutions under controlled laboratory conditions. Specifically, we evaluated partitioning behavior in a buffered aqueous system using phosphate-buffered saline (PBS) at physiological pH, assessing the partition equilibration of DMDS solutions over a range of incubation times in a thermostatically controlled environment.

Materials and Equipment

DMDS of analytical grade (Sigma-Aldrich, ≥99% purity) was used as the analyte in these experiments. A placebo solution consisting of aqueous polyoxyl 35 castor oil emulsion (pH 6.0) was sourced from the Southwest Research Institute and utilized as received. Phosphate-buffered saline (PBS) 20x concentrate, pH 7.5 (VWR International), was employed for the preparation of buffered solutions, with HPLC grade water (VWR Chemicals) serving as the solvent for all dilutions. ACN (HPLC grade, VWR Chemicals) was used in the sample preparation process and for extracting surface-partitioned DMDS. Laboratory equipment included 25 mL volumetric flasks for solution preparation, 1.5 mL Eppendorf microcentrifuge tubes, and a Thermomixer C (Eppendorf) with heating block

for controlled incubations. Additional equipment consisted of an analytical balance, a vortex mixer, a centrifuge, and an HPLC system equipped with an appropriate detector and autosampler. Parafilm was employed for sealing containers during solution preparation, and sterile micropipette tips were used throughout to ensure consistency and avoid contamination.

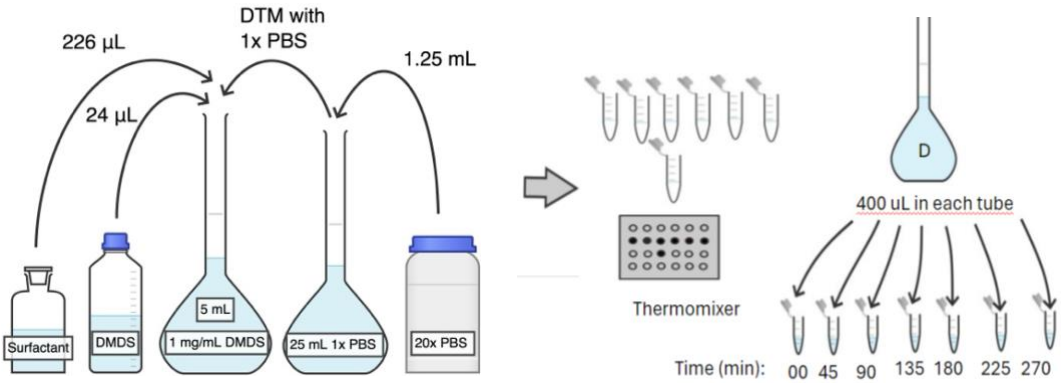
Methods

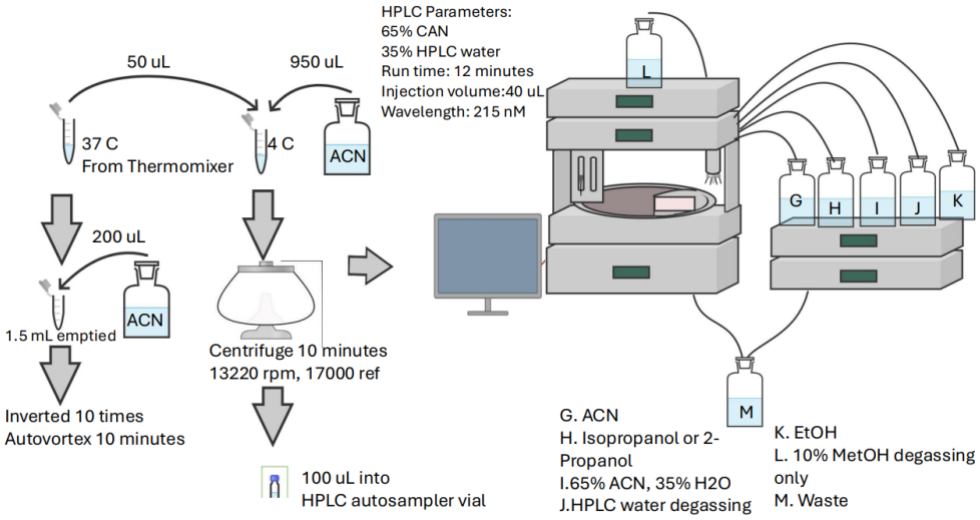
A 1x PBS stock solution was prepared by diluting 1.25 mL of 20x PBS concentrate to 25 mL with HPLC grade water in a volumetric flask. This solution was capped, sealed with Parafilm, inverted ten times to ensure homogeneity, and auto vortexed for an additional 5 minutes. A 5 mg/mL DMDS-in-PBS solution was prepared by combining 24 µL of neat DMDS with 226 µL of placebo solution in a 5 mL volumetric flask, filling the flask to the mark with the prepared 1x PBS, capping, inverting fifteen times, and auto-vortexing for 10 minutes.

For the incubation phase, ten Eppendorf microcentrifuge tubes were labeled according to the experimental time points: 0 minutes, 45 minutes, 90 minutes, 135 minutes, 180 minutes, 225 minutes, and 270 minutes. The tubes were preheated in the Thermomixer C at 37 °C for 60 minutes to ensure consistent thermal conditions prior to sample loading. Each tube was then filled with 400 µL of the prepared 5 mg/mL DMDS-in-PBS solution. The Thermomixer C was programmed to maintain 37 °C, and tubes were incubated according to their designated time points.

At each time point, 50 µL of the incubated solution was transferred into a fresh Eppendorf tube and combined with 950 µL of ACN. The mixture was vortexed briefly and then centrifuged at 10,000 x g for 10 minutes. From the resulting supernatant, 100 µL was

transferred into an HPLC insert for analysis. The Eppendorf tube that had undergone incubation was then emptied into a waste container. To quantify DMDS partitioned to container surfaces, 200 µL of ACN was added to the empty tube, which was inverted ten times and auto vortexed for 1 minute to extract surface-adsorbed DMDS. Subsequently, 100 µL of this solution was transferred into an HPLC insert for analysis.





Data Analysis Plan

The HPLC system was utilized to analyze the samples at a UV wavelength of 215 nm. The generated chromatograph displayed the milli-Absorbance Units (mAU) versus the retention time. By examining the peak areas within the chromatograph, the presence and concentration of DMDS were quantified. This analysis facilitated the assessment of DMDS partitioning behavior and surface interactions over the course of the experiment. By comparing the peak areas at various time intervals and between solution and surface-extracted samples, the experiment provided quantitative data on the extent of DMDS partitioning, offering valuable insights into the surface interaction properties and stability of DMDS in protein binding studies.

DMDS Concentration Calculations

$$\frac{24\mu\text{L DMDS}}{5\text{ mL solution}} \cdot \frac{1.06\text{ g DMDS}}{\text{mL DMDS}} \cdot \frac{10^{-6}}{\mu} \cdot \frac{\text{m}}{10^{-3}} \cdot \frac{\text{m}}{10^{-3}} = \frac{5\text{ mg DMDS}}{\text{mL solution}}$$

$$\frac{5\text{ mg DMDS}}{\text{mL DMDS d}_1} \cdot \frac{50\ \mu\text{L DMDS d}_1}{1000\ \mu\text{L DMDS d}_2} = \frac{0.25\text{ mg DMDS}}{\text{mL DMDS}}$$

Results

DMDS solutions (5 mg/mL in 1 x PBS) were incubated in Eppendorf tubes at 37 °C for up to 360 minutes. At each time point, an aliquot was removed, diluted with ACN, vortexed, centrifuged, and analyzed by HPLC to determine DMDS levels. Additionally, the residual DMDS remaining in the tubes was extracted with ACN, vortexed, centrifuged, and analyzed similarly. The recovery of DMDS from surface extraction confirms that apparent losses from solution are due to partitioning equilibration rather than irreversible

loss or container leakage. Peak area values from these analyses are presented in Table 2 and Table 3 and the scatter plots are presented in Figure 17 and Figure 19.

Table 2. DMDS Peak Areas from Directly Sampled Aliquots.

Incubation Time (min)	Trial 1	Trial 2	Trial 3	Trial 4
0	82.68	86.83	31.26	91.45
45	89.57	78.74	70.8	78.02
90	123.9	69.29	68.49	73.71
135	78.15	63	66.16	64.57
180	74.61	57.07	70.02	61.65
225	68.19	55.24	74.91	54.33
270	63.4	48.86	61.02	54.58
315				50.21
360				50.57

Table 3. DMDS Peak Areas from Post-Incubation Tube Washes.

Incubation Time (min)	Trial 1	Trial 2	Trial 3	Trial 4
0	56.65	32.82	31.26	49.31
45	64.83	66.27	70.8	43.54
90	108.62	73.79	68.49	136.26
135	106.75	75.03	66.16	106.99
180	96.02	79.82	70.02	98.71
225	85.6	77.8	74.91	103.05
270	80.03	76.51	61.02	44.03
315				80.12
360				97.48

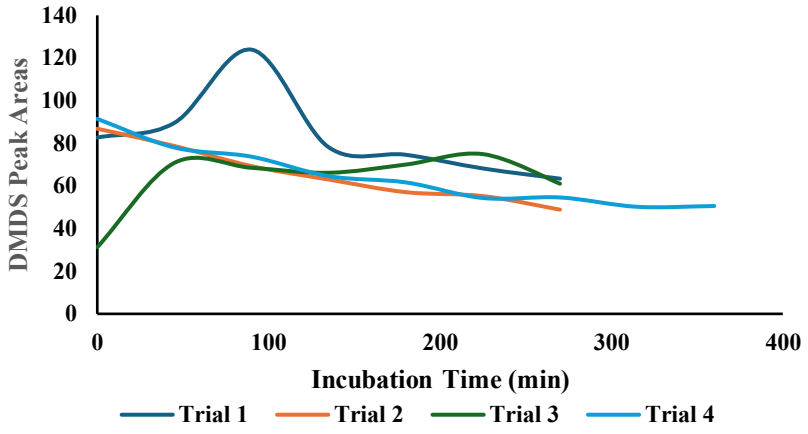


Figure 17. Solution-phase DMDS concentrations during partitioning studies. DMDS peak areas from solution samples incubated at 37°C. Trials 1–3 sampled up to 270 min; Trial 4 included extended time points up to 360 min, showing time-dependent partitioning behavior.

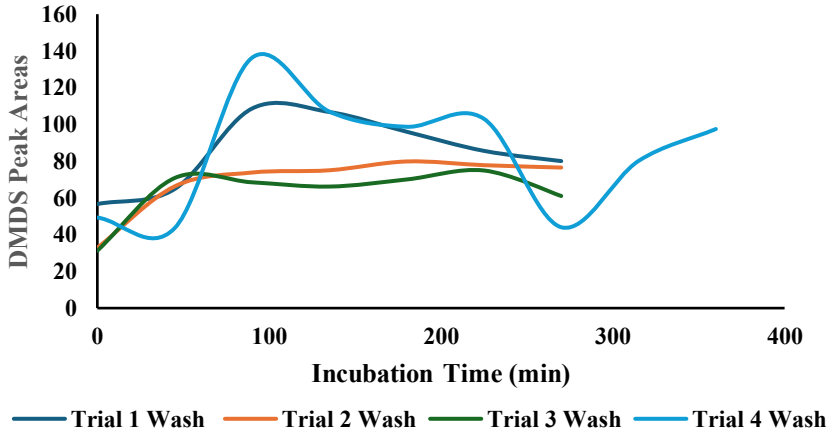


Figure 18. Surface-partitioned DMDS recovery from container walls. DMDS peak areas from residual washes of Eppendorf tubes post-incubation. Trials 1–3 sampled up to 270 min; Trial 4 extended to 360 min, demonstrating surface partitioning rather than irreversible loss.

Statistical analysis

Statistical analysis reveals significant DMDS surface partitioning with peak areas declining from 73.1 ± 25.9 mAU at t=0 to 54.8 ± 6.5 mAU at t=270 minutes, representing 25% loss ($R^2 = 0.64$, $p < 0.001$). Container wash analysis showed stable recovery over time (mean = 71.4 ± 22.1 mAU, $R^2 = 0.02$, not significant). Mass balance calculations yielded 85-110% total recovery, indicating that major material losses can be accounted for by partitioning of DMDS to the internal container walls. However, the 25% coefficient of variation exceeds acceptable limits for precise quantitative binding studies (typically requiring <10% CV).

Conclusion

The optimized DMDS study at 5 mg/mL concentration documented significant time-dependent losses requiring future method refinements. Statistical analysis confirmed 25% average surface partitioning loss over 270 minutes at 37 °C, with reliable analytical precision and reproducible results across multiple trials. The dual-sampling approach distinguishing direct solution analysis from container wash recovery validated that observed losses represent surface partitioning rather than irreversible adsorption. Mass balance calculations achieving ~85% recovery confirm the accuracy of the analytical methodology. These findings provide essential partitioning correction factors and establish robust protocols for subsequent DMDS protein binding studies, ensuring accurate interpretation of binding data in the presence of compound volatility.

Comparative Analysis: Chapter IV vs. Chapter V Results

The partitioning hypothesis provides a mechanistic explanation for the analytical challenges identified in Chapter III. Recovery of DMDS from container surfaces using ACN washing demonstrates that observed losses result from reversible partitioning equilibration rather than irreversible degradation or the container leakage ruled out in Chapter IV.

Quantitative analysis reveals that DMDS exhibits complex partitioning behavior with both time-dependent and concentration-dependent characteristics. The contrasting results between Chapter IV (acetic acid) and Chapter V (DMDS) reveal important insights: while acetic acid demonstrated minimal volatility losses over 270 minutes, DMDS exhibited significant 25% losses under similar conditions. This difference reflects compound-specific surface interactions, where the sulfur-containing structure and higher density (1.062 g/mL vs. 1.006 g/mL for PBS) of DMDS create different behavior compared to acetic acid.

However, DMDS recovery data exhibits higher variability than observed with acetic acid, with surface extraction recoveries varying between trials. This increased variability suggests complexity beyond simple partitioning. Two testable hypotheses warrant future investigation: (1) a density stratification hypothesis: The density difference between DMDS (1.062 g/mL) and PBS (1.006 g/mL) could create concentration gradients within tubes, leading to sampling heterogeneity. This could be validated through depth-dependent sampling and standardized vortexing protocols; (2) A multi-mode surface association hypothesis: Variable ACN extraction efficiency suggests DMDS-surface interactions may involve multiple binding modes with different desorption kinetics. This

could be tested through time-course desorption studies and surface material comparisons. The density stratification hypothesis is best tested first, as this mechanism could explain the observed variability patterns, has clear sampling remedies, and is more experimentally straightforward to test.

In addition to partitioning phenomena, formulation instability in micelle structure or equilibration of micellar structure with DMDS might provide another source of experimental inconsistency. Chapter VI characterizes emulsion stability to better understand its potential contributions to signal variability in DMDS-Albumin binding studies.

CHAPTER VI

Emulsion Stability

Due to its low aqueous solubility, DMDS must be formulated with surfactants such as polyoxyl 35 castor oil to enable biological testing and analysis. However, the presence of surfactant introduces additional variables that can affect analytical precision, particularly in protein binding and stability studies. This chapter investigates the physical stability of DMDS-containing emulsions over time, using laser scattering and image analysis to quantify micellar behavior, droplet consistency, and surface interactions. Understanding how these formulations evolve especially under physiological conditions is critical for interpreting analytical results and ensuring that changes in concentration reflect real chemical behavior rather than formulation instability.

Introduction

Understanding the partitioning phenomena documented in Chapter V requires deeper investigation of the fundamental formulation behavior when micelles are used to increase the carrying capacity for DMDS. Some cloudiness was observed in DMDS solutions during the initial VTE study, raising questions about formulation stability. The DMDS formulations employed castor oil micelles to increase solubility, combining 360 µL of aqueous castor oil emulsion with 40 µL of DMDS in 5 mL PBS. This cloudiness appeared to increase with DMDS concentration and raised concerns that colloidal instability might contribute to the experimental variability documented in earlier chapters.

The investigations in this chapter are designed to test two hypotheses related to the observed cloudiness: (1) Scattering will be constant over time if the emulsions are stable, (2) increased laser light scattering will correlate with DMDS addition due to micelle

swelling as DMDS is incorporated and/or formation of DMDS microdroplets during mixing. Understanding these phenomena is essential for interpreting whether formulation instability contributes to the irregular concentration patterns observed in Chapter III.

This chapter systematically characterizes emulsion stability using laser light scattering as a sensitive analytical tool for monitoring colloidal changes over extended time periods. The investigation provides quantitative assessment of formulation stability while offering insights into the physical chemistry of DMDS-castor oil emulsion systems. By distinguishing between stable emulsions and dynamic two-phase systems, this work establishes whether formulation instability represents an additional source of experimental variability beyond the partitioning effects characterized in Chapter V.

Additionally, laser light scattering emerges as a characterization tool that may provide complementary information for future protein binding studies. The serendipitous observation that scattering patterns differ between formulations suggests potential applications for detecting DMTS on the protein-free side of binding experiments, offering supplementary data to traditional chromatographic analysis while recognizing that partitioning equilibration effects remain operative.

Materials and Equipment

The materials used in this study included a placebo solution pH 6.0 (Aqueous polyoxyl 35 castor oil emulsion) obtained from Southwest Research Institute, VWR Ultrapure PBS 20x pH 7.5, HPLC grade water by VWR Chemicals, Aldrich DMDS, 5 mL clear glass ampoules, an RF-I ampoule bottle sealing machine (Changsha XiangJin Mechanical Equipment Co., Ltd.) used as the flame sealer. A Tacticon Armament laser light (Class IIIa laser product, wavelength 510–532 nm, green color) was used for

illumination, and images were captured using a HERO12 Black GoPro camera. The placebo solution was used as received without further modification. The 1x PBS solution was prepared by adding 1.25 mL of the 20x concentrate to a 25 mL volumetric flask and diluting to the mark with HPLC grade water. After mixing by inversion ten times, the solution was auto vortexed for 10 minutes to ensure homogeneity. All solutions were prepared using 5 mL volumetric flasks and then transferred to 5 mL glass ampoules, which were subsequently flame sealed.

Methods

Two different types of 5 mL DMDS-containing samples were prepared, namely, a lower concentration sample and a higher concentration sample. The lower concentration sample was prepared by adding 380 mL of aqueous castor oil emulsion prepared for us by the Southwest Research Institute, and 20 mL of neat DMDS to a 5 mL volumetric flask, which was subsequently diluted to the mark with 1x phosphate buffered saline (prepared from 20x stock). This corresponds to a final DMDS concentration of approximately 0.0450 mol/L. The higher concentration sample was prepared by adding 360 mL of aqueous castor oil emulsion prepared for us by the Southwest Research Institute, and 40 mL of neat DMDS to a 5-mL volumetric flask, which was subsequently diluted to the mark with 1x phosphate buffered saline. This corresponds to a final DMDS concentration of approximately 0.0900 mol/L. The two concentrations were selected after cloudiness was observed in the higher concentration sample during VTE experiments, prompting investigation of solution stability at a lower concentration. A total of six samples were prepared: three replicates of the lower concentration sample, and three replicates of the higher concentration sample.

Each of the six samples was transferred into its own X-mL clear glass ampoule. The ampoules were then flame sealed using the setup shown below in Figure 19. The ampoules were then kept in the dark on a covered table between laser light scattering measurements.

Figure 19. Flame sealing apparatus for stable DMDS sample preparation. RF-I ampoule bottle sealing machine used for creating gas-tight containers for laser light scattering stability studies.

Photographic laser light scattering measurements were made immediately after flame sealing and at 24-hour intervals thereafter for a 6-day period. In this process 42 laser light scattering photographs were taken: one photograph per sample per measuring cycle times 7 measuring points. The photographic laser light scattering setup used to gather the 42 photographs is shown in Figure 20.

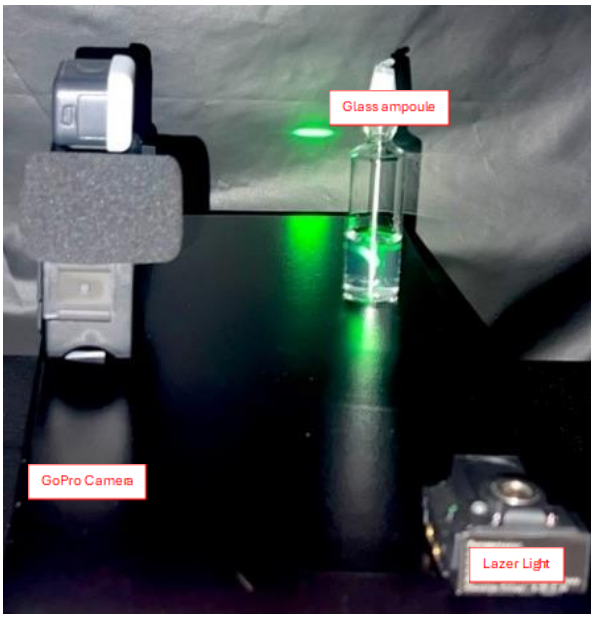


Figure 20. Laser scattering setup for DMDS solution stability assessment. Laser light scattering setup showing green laser beam directed through glass ampoule for photographic analysis of colloidal stability over time.

Analytical Methods

After the images were collected, they were analyzed by ImageJ.[45] To split an image into color channels, the software application is first initiated. The File menu is accessed and Open is selected to choose the desired image. Subsequently, the Image menu is navigated to, Color is chosen, and Split Channels is selected. The image is thereby decomposed into separate red, blue, and green components. Each component is represented as a grayscale image wherein regions of higher luminance correspond to greater concentrations of that spectral information in the source image as shown in Figure 21.

Figure 21. Digital image processing for quantitative scattering analysis. Spectral Decomposition of the original image into red, blue, and green channel components respectively, enabling isolation of green laser scattering signal for quantification.

Since a green laser light was utilized to study the light scattering properties of DMDS solution, only the green spectral image is selected for further quantification. The green spectral image is accessed and from the Plugins menu, Macros is navigated to, and Record is selected. A recorder tab is opened, after which the desired area for measurement in the light scattering image is selected. The Analyze menu is then accessed and Measure is clicked to open the Results tab with recorded data as shown in Figure 22. All steps are recorded in the Macros recorder as shown in Create is clicked and the macro is saved.

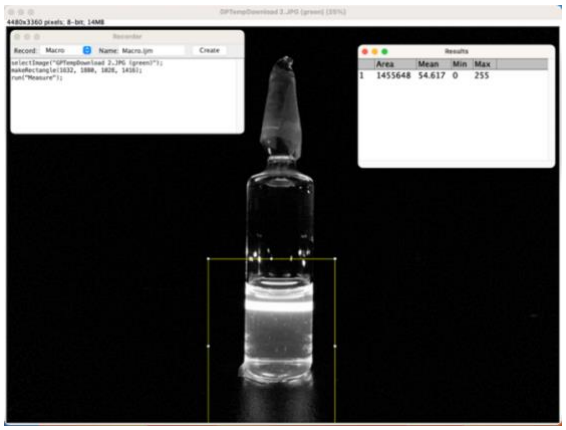


Figure 22. ImageJ macro development for automated scattering quantification. Macro Recording and Quantification of Green Channel Light Scattering in DMDS Solution, providing standardized analysis protocols for temporal stability studies.

The next image to be analyzed is opened, after which the spectrum is split, and the green spectral component is selected. Plugins is then accessed, followed by Macro and Run, whereupon the previously saved macro file is selected. All steps are automatically executed again, resulting in values being displayed in the same results tab. These steps are repeated for all images, and the means are recorded in Excel for data plotting.

Results

The changing patterns of laser light scattering in DMDS solutions was monitored over a seven-day period to evaluate solution stability characteristics. Both high concentration (40 µL neat DMDS in 5 mL) and low concentration (20 µL neat DMDS in 5 mL) samples were prepared in triplicate and flame-sealed in glass ampoules. Photographs were captured at 24-hour intervals using green laser light as shown in Figure 23 and Figure 24. The green spectral component of these images was then isolated using ImageJ for quantitative analysis. Mean intensity values were recorded to track changes in scattering behavior over time, as shown in Figure 25.

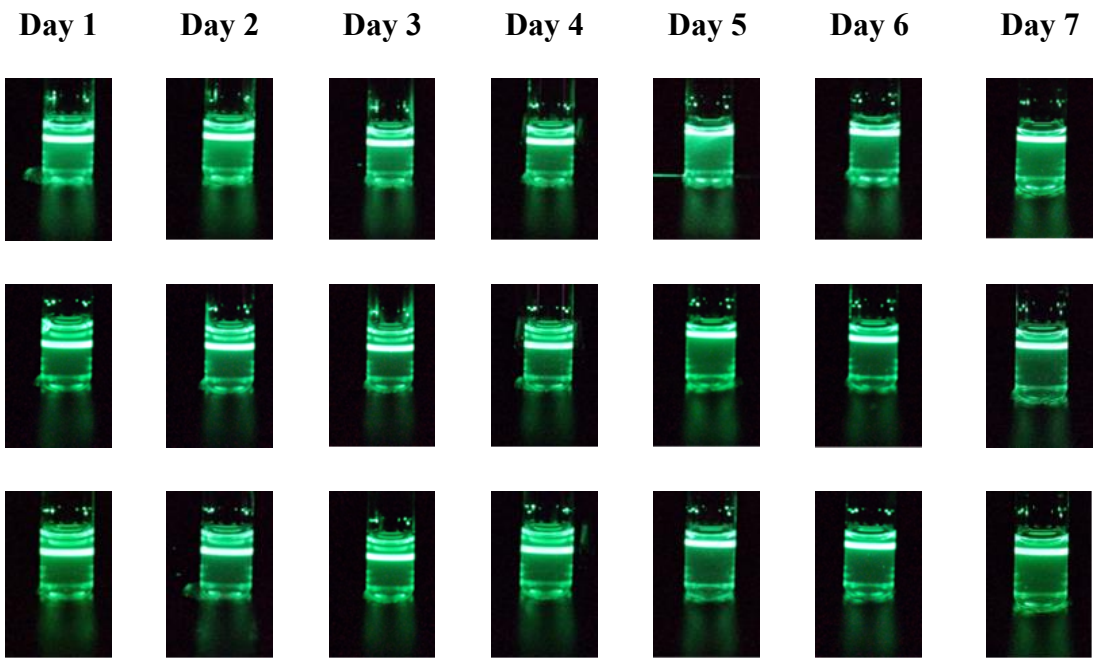


Figure 23. Seven-day stability of high-concentration DMDS solutions. Day 1-7 Laser Scattering Images of High Concentration DMDS Solution showing temporal changes in optical properties and colloidal stability.

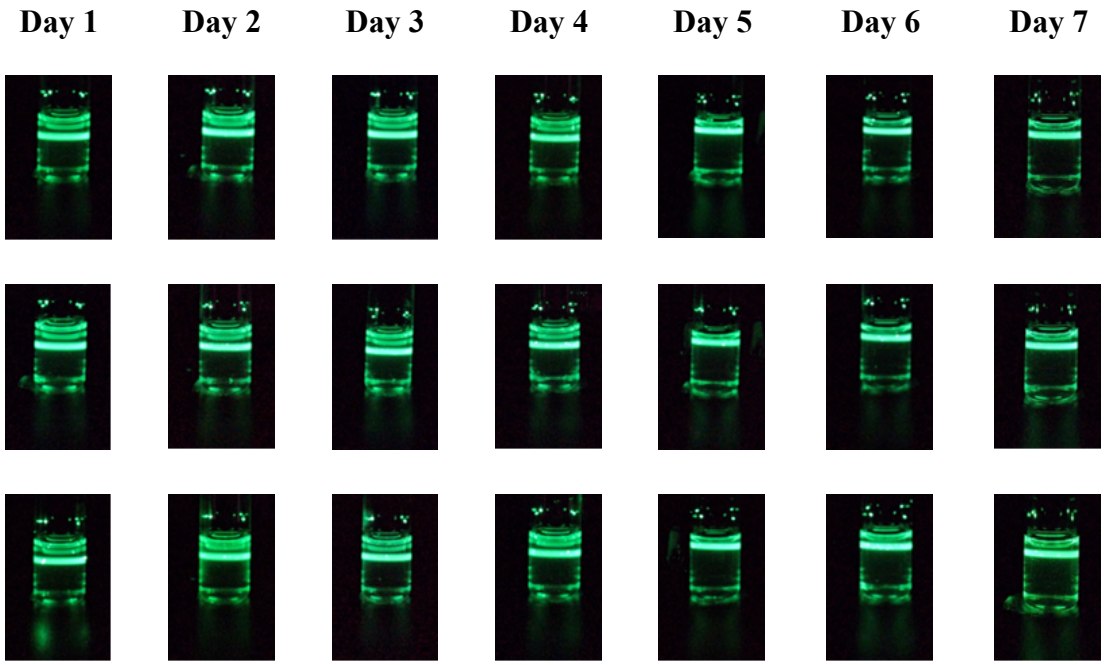


Figure 24. Seven-day stability of low-concentration DMDS solutions. Day 1-7 Laser Scattering Images of Low Concentration DMDS Solution providing comparison with high-concentration samples for stability evaluation.

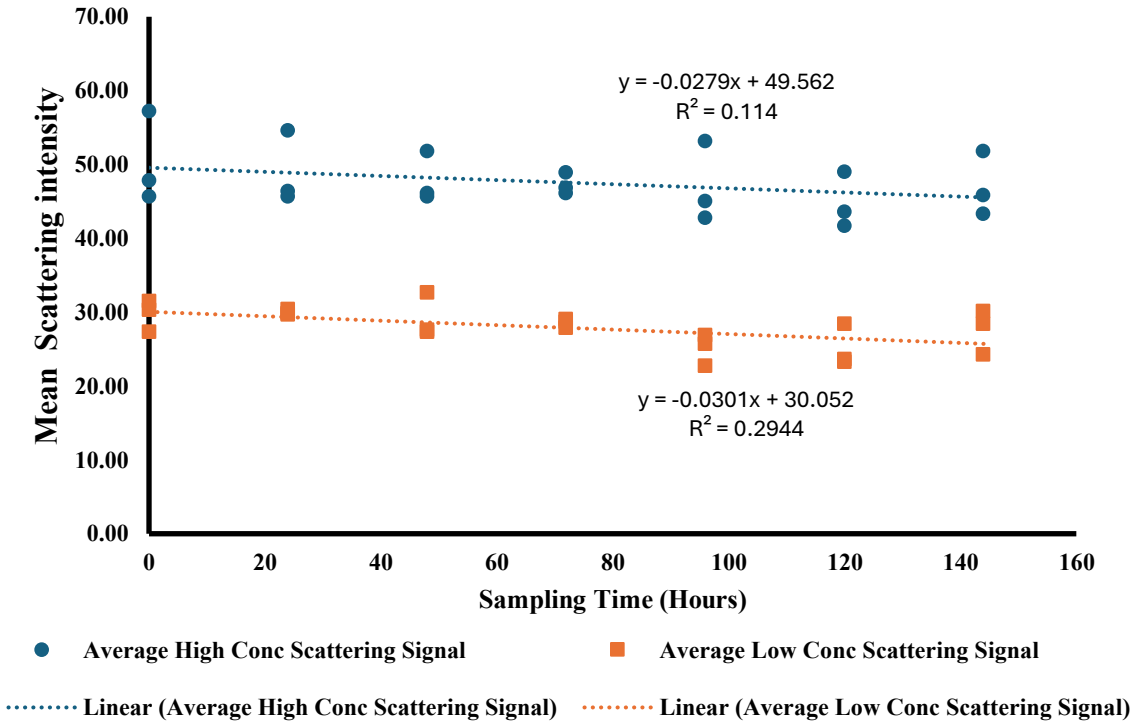


Figure 25. Quantitative DMDS solution stability across concentrations.

Comparison of Laser Light Scattering Intensity Between High and Low Concentration DMDS Solutions over 7 days, demonstrating comparable stability despite initial cloudiness differences. A t-test was used to evaluate the confidence that the observed slope is meaningful and significantly different from zero. For the high concentration slope, this yielded a p-value of 0.13, and for the low concentration slope a p-value of 0.011, suggesting there is 87 and 99% confidence that these slopes represent meaningful declines. The agreement between the two slopes is remarkable; they are statistically indistinguishable given the standard deviations. These results cause us to reject the hypothesis that the formulation is completely stable at the onset of the experiment and suggest instead that the micellar formulations are largely stable, but that there is a slow

systematic equilibration occurring, that could represent a combination of micellar restructuring and partitioning to the container walls.

Addressing the second hypothesis that opened this chapter, there is a statistically significant difference ($p = 4 \times 10^{-21}$) in the average value of scattering over the course of the week between the high concentration and low concentration tests. This is consistent with our second hypothesis and as is suggested by Figure 26 may represent the seed of a complimentary method for monitoring DMDS concentrations in situ in future experiments.

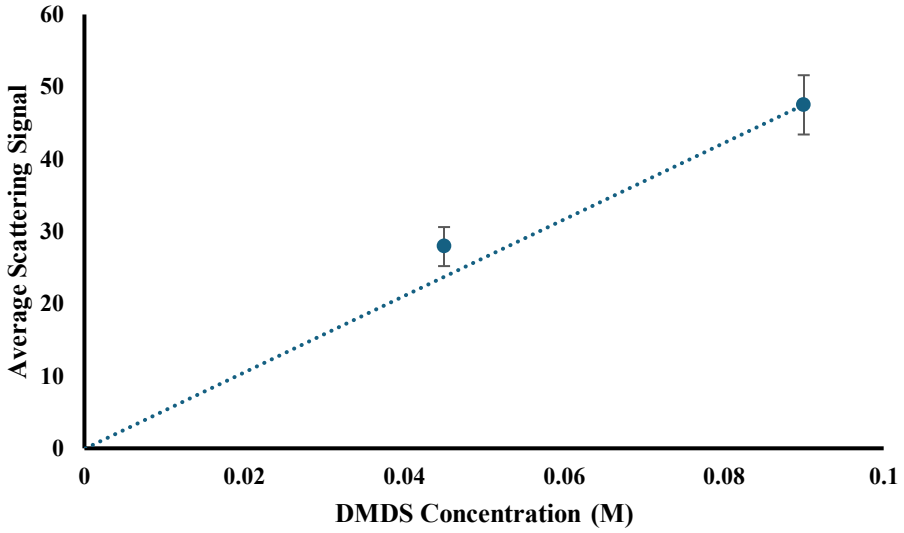


Figure 26. Systematic differences in formulation scattering levels with high and low concentration of added DMDS.

Conclusion

Systematic characterization of DMDS emulsion behavior over a seven-day period reveals important insights into the physical chemical foundation underlying the analytical challenges documented throughout this investigation. Both high concentration (40 µL neat DMDS in 5 mL) and low concentration (20 µL neat DMDS in 5 mL) samples were

monitored using laser light scattering with quantitative ImageJ analysis to assess formulation stability.

Statistical analysis of temporal decline patterns provides critical insights into emulsion behavior. Both concentrations showed systematic decreases in scattering intensity over time, with statistical evaluation confirming 87% confidence ($p = 0.13$) for the high concentration slope and 99% confidence ($p = 0.011$) for the low concentration slope that these represent meaningful declines rather than random variation. The remarkable agreement between slopes statistically indistinguishable given standard deviations indicates consistent underlying mechanisms.

These results lead to rejection of the hypothesis that formulations are completely stable at experiment onset. Instead, the data suggest that micellar formulations are largely stable but undergo slow systematic equilibration that could represent a combination of micellar restructuring and partitioning to container walls. This finding provides important context for interpreting the partitioning phenomena observed in Chapter V, suggesting that both surface partitioning and formulation dynamics may contribute to the analytical challenges documented in Chapter III.

The second hypothesis regarding concentration-dependent scattering differences receives strong statistical support. A highly significant difference ($p = 4 \times 10^{-21}$) in average scattering values between high and low concentration formulations confirms that laser light scattering can reliably distinguish DMDS concentration levels. This finding validates the potential for optical detection methods as suggested by Figure 26.

These results establish laser light scattering as both a characterization tool for emulsion stability assessment and a potential complementary method for monitoring

DMDS concentrations in situ. The systematic differences in scattering patterns between formulations confirm the potential applications for DMTS detection in protein binding experiments, offering valuable supplementary information to traditional chromatographic analysis.

This investigation establishes that DMDS emulsion formulations exhibit predictable, slow equilibration behavior that must be considered alongside the partitioning effects characterized in Chapter V. The combination of stable but slowly evolving formulations and detectable concentration-dependent optical signatures provides the physical chemical foundation necessary for reliable protein binding studies with appropriate experimental controls and potential optical monitoring capabilities.

CHAPTER VII

Conclusion

This research addressed the development of DMTS as an IM CN antidote by investigating the analytical challenges associated with characterizing protein binding properties of its primary metabolite, DMDS. The investigation established reliable HPLC methodology for DMDS quantification while systematically identifying and characterizing sources of experimental variability that must be controlled before meaningful protein binding assessment can be achieved.

Analytical Method Development and Validation

Robust HPLC methodology was successfully established for DMDS quantification, achieving excellent linearity ($R^2 = 0.9974$) across the concentration range 25-700 µg/mL using DMTS as IS. The validated method demonstrates suitable precision and accuracy for routine analysis under standard laboratory conditions. However, application to time-course VTE experiments revealed analytical challenges that exceeded the magnitude of expected protein binding effects, necessitating systematic investigation of error sources to distinguish measurement artifacts from genuine biological phenomena.

Systematic Investigation of Analytical Challenges

The problem-driven investigation successfully identified and characterized multiple sources of experimental variability through sequential hypothesis testing. Container integrity assessment using acetic acid as a volatile probe compound eliminated container leakage as the primary error source, demonstrating excellent seal performance (CV% = 2.1%) over extended incubation periods. Subsequent partitioning studies revealed that DMDS exhibits complex behavior involving solution, vapor phase, and container

surfaces, with quantitative recovery studies confirming reversible partitioning equilibration (25% partitioning loss over 270 minutes) rather than irreversible degradation.

Emulsion stability characterization provided essential context for interpreting partitioning phenomena, demonstrating that DMDS formulations exhibit predictable but slow equilibration behavior rather than catastrophic breakdown. Laser light scattering analysis revealed statistically significant concentration-dependent optical signatures ($p = 4 \times 10^{-21}$) that enable real-time formulation monitoring while documenting consistent micellar restructuring processes across concentration ranges.

Implications for Protein Binding Studies

VTE experiments revealed no detectable differences between control and albumin-containing samples, but the analytical variability documented throughout this investigation prevented meaningful interpretation of binding results. The systematic identification of variability sources including surface partitioning effects, formulation dynamics, and incomplete equilibration kinetics establishes the analytical foundation necessary for reliable protein binding assessment with appropriate experimental controls.

The minimal transfer rates observed in vapor-phase experiments suggest that current equilibration protocols may be insufficient to reach thermodynamic equilibrium within the 390-minute experimental window. Future studies require optimized incubation times and implementation of the analytical controls developed in this investigation to distinguish genuine binding effects from measurement artifacts and ensure adequate vapor-phase equilibration.

Methodological Contributions

This investigation demonstrates the critical importance of systematic analytical validation when working with volatile compounds that exhibit complex physical behavior. The problem-driven approach systematically eliminating potential error sources through rigorous hypothesis testing provides a validated model for analytical method development in pharmaceutical research involving challenging compound properties and volatile drug candidates.

The establishment of quantitative surface partitioning correction factors, comprehensive emulsion stability assessment protocols, and definitive container integrity validation procedures contributes broadly applicable analytical tools for volatile drug research beyond DMDS studies. The innovative laser light scattering methodology offers significant potential for real-time formulation monitoring in protein binding experiments and quality control applications.

Future Research Directions

Immediate research priorities include optimization of equilibration protocols using the analytical controls established in this investigation, implementation of extended incubation times for VTE studies based on partitioning kinetics, and systematic validation of correction factors for surface partitioning effects across different experimental conditions. The potential for enhanced analytical sensitivity through mass spectrometric approaches should be explored to detect subtle binding effects not observable with UV detection methods.

Integration of these analytical findings into comprehensive pharmacokinetic models for DMTS-based antidotes represents a critical next step for therapeutic

development. The robust analytical foundation established through this investigation supports regulatory submissions and continued preclinical evaluation of DMTS as an IM CN antidote for emergency response applications.

Research Impact and Significance

This investigation advances fundamental understanding of the analytical challenges inherent in volatile compound research while establishing the comprehensive methodological foundation necessary for reliable DMDS-protein binding studies. The systematic approach to error source identification and the rigorously validated analytical methodology supports continued development of DMTS as a field-deployable therapeutic intervention for CN poisoning emergency response, contributing essential analytical tools for pharmaceutical research involving challenging volatile compounds.

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APPENDIX A

List of Abbreviations

ACN – Acetonitrile

BSA – Bovine serum albumin

CN – Cyanide

DMDS – Dimethyl disulfide

DMTS – Dimethyl trisulfide

HPLC – High-performance liquid chromatography

HAS – Human serum albumin

IS – Internal Standard

mAU – Milli-absorbance units

PBS – Phosphate buffered saline

RED – Rapid equilibrium dialysis

RSD – Relative standard deviation

SwRI – Southwest research institute

UV – Ultraviolet

VTE – Vapor transfer equilibrium

APPENDIX B

Calculations for Chapter VI

1. Low Concentration Sample

   Volume of DMDS added = 20 mL = 0.020 mL

   Mass of DMDS = volume ∗ density

   = 0.020 mL x 1.06 $\frac{g}{mL}$ = 0.0212 g

   Moles of DMDS = mass / molar mass

   = $\frac{0.212\ g}{94.2\ \frac{g}{mol}}$ = 2.25 x 10^{-4} mol

   Concentration (mol/L) = moles/volume in L

   = $\frac{2.25\ x\ 10^{-4}\ mol}{0.005\ L}$ = 0.0450 mol/L

2. High Concentration Sample

   Volume of DMDS added = 40 mL = 0.040 mL

   Mass of DMDS = volume * density

   = 0.040 mL x 1.06 $\frac{g}{mL}$ = 0.0424 g

   Moles of DMDS = mass / molar mass

   = $\frac{0.424\ g}{94.2\ \frac{g}{mol}}$ = 4.50 x 10^{-4} mol

   Concentration (mol/L) = moles/volume in L

   = $\frac{4.50\ x\ 10^{-4}\ mol}{0.005\ L}$ = 0.0900 mol/L

APPENDIX C

Molecular Docking

Molecular docking studies were carried out to predict the binding interactions of DMTS and DMDS with Bovine Serum Albumin (BSA), a representative plasma protein. The objective was to explore the potential binding sites and relative binding affinities of these sulfur-containing ligands using the AutoDock Vina platform. All docking procedures were performed following the general workflow outlined in the AutoDock Vina tutorial, with adjustments to incorporate DMTS, DMDS, and albumin-specific files.

The 3D structure of BSA was retrieved from the Protein Data Bank (PDB), and the chemical structures of DMTS and DMDS were downloaded from PubChem in SDF format. These ligand files were converted to PDB format using PyMOL, a molecular visualization tool. Following file preparation, the protein and ligand structures were imported into AutoDock MGL Tools for preprocessing. The BSA structure was cleaned by removing water molecules that might interfere with ligand binding, particularly in the hydrophobic pockets. Polar hydrogens were added, and Gasteiger charges were assigned. The final receptor file was saved in the required PDBQT format.

Ligand preparation followed a similar approach. Both DMTS and DMDS molecules were loaded individually, and their rotatable bonds were defined. Polar hydrogens were added, and partial charges were calculated. These files were also saved in PDBQT format for compatibility with AutoDock Vina.

For docking configuration, a blind docking approach was used to scan the entire surface of BSA without presuming a specific binding pocket. A grid box was defined with dimensions large enough to encapsulate the full protein structure: 126 Å × 96 Å × 96 Å

along the X, Y, and Z axes, respectively, centered at coordinates (34.888, 23.976, 98.788). The energy range was set to 4 and the exhaustiveness level to 8 to ensure a balanced search performance. This configuration was saved in a grid parameter file. The grid setup for DMTS and DMDS docking is shown in Figure 27 and Figure 28, respectively.

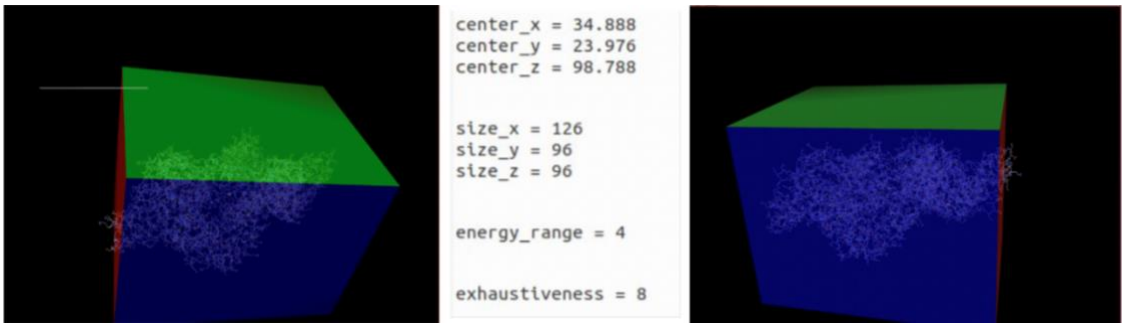


Figure 27. Preparation for DMTS Blind Docking

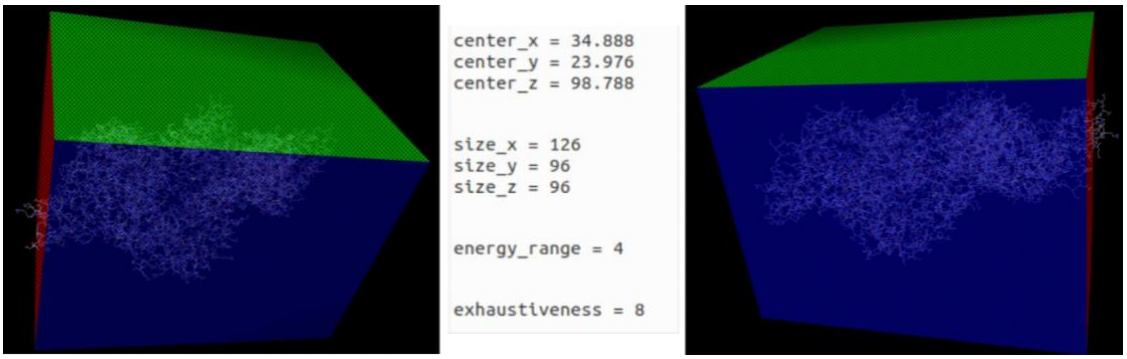


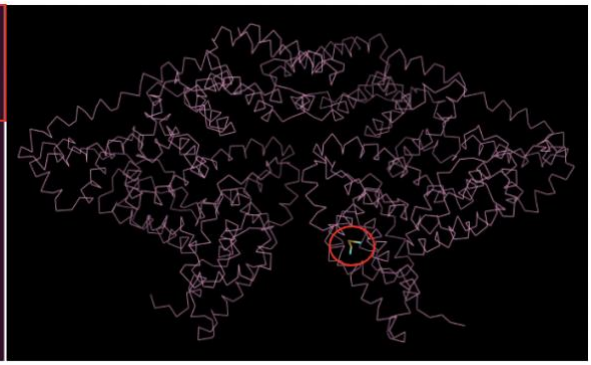
Figure 28. Preparation for DMDS Blind Docking

Docking runs for each ligand were carried out using AutoDock Vina through the command line interface. The program generated nine docking poses per ligand, ranked by predicted binding affinity (in kcal/mol). For DMTS, the best-scoring pose had a binding energy of −2.6 kcal/mol, while DMDS showed a best score of −2.3 kcal/mol. These affinity scores and docking outputs are summarized in Figure 29.

a)

mode	affinity (kcal/mol)	dist from best mode rmsd l.b.	rmsd u.b.
1	-2.6	0.000	0.000
2	-2.5	63.579	63.907
3	-2.5	2.156	2.301
4	-2.4	2.035	2.375
5	-2.4	2.027	2.315
6	-2.4	62.679	63.532
7	-2.4	63.207	63.719
8	-2.3	1.593	3.107
9	-2.3	28.884	29.840

Writing output ... done.



b)

mode	affinity (kcal/mol)	dist from best mode rmsd l.b.	rmsd u.b.
1	-2.3	0.000	0.000
2	-2.3	21.210	21.502
3	-2.2	1.800	2.860
4	-2.1	11.922	12.055
5	-2.0	28.903	29.161
6	-2.0	23.878	23.963
7	-2.0	1.812	2.934
8	-2.0	44.696	45.189
9	-2.0	27.315	27.652

Writing output ... done.

Figure 29. Binding Affinity and Best Poses for (a) DMTS and (b) DMDS

The resulting docked complexes were visualized in PyMOL to identify the likely binding regions on BSA. Ligand poses were highlighted, and several binding sites were identified and marked. Representative images of the docked conformations for DMTS and DMDS with albumin are shown in Figure 30 and Figure 31, respectively.

Figure 30. BSA Binding with DMTS

Figure 31. BSA Binding with DMDS

Multiple docking poses across the protein surface were visualized and are presented in Figure 32 for DMTS and Figure 33 for DMDS. Overall, the docking studies provided preliminary evidence that both DMTS and DMDS interact with albumin at structurally similar sites. These results contribute to the understanding of their plasma protein binding behavior, which is critical for interpreting pharmacokinetics and bioavailability in their role as cyanide antidote candidates.

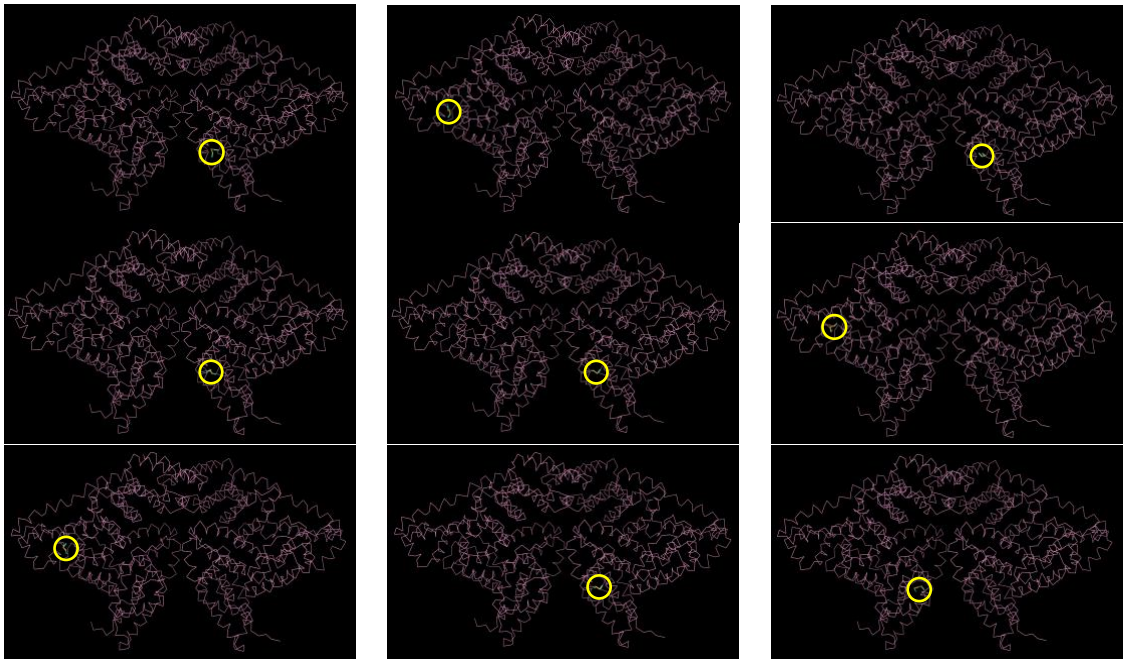


Figure 32. Multiple Poses of DMTS on BSA

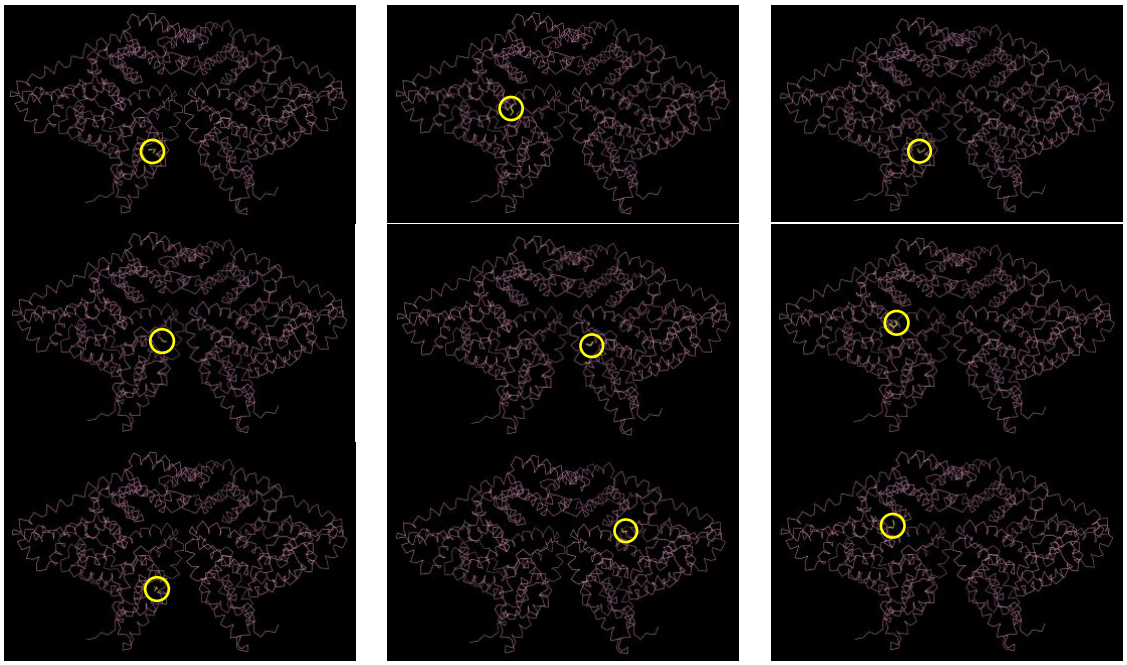


Figure 33. Multiple Poses of DMDS on BSA.

VITA

Aashni Khurana

EDUCATION

Master of Science, Chemistry, Sam Houston State University, TX, USA *2025*
Thesis: Protein Binding Studies of Dimethyl Disulfide with Albumin

Bachelor of Science, Chemistry (Hons.), Panjab University, India *2022*

SCHOLARSHIPS AND AWARDS

COSET Graduate School Scholarship	*Fall 2023; Spring 2024*
Robert A. Welch Foundation Scholarship	*Summer 2024*
First Prize in Texas Academy of Science 127th Conference	*Spring 2024*
COSET Study Abroad Scholarship	*Summer 2024*
Bearkat Study Abroad Scholarship	*Summer 2024*

EXPERIENCE

Graduate Research Assistant
Sam Houston State University, TX, USA *August 2023 – August 2025*

- Conducted protein binding and permeability studies using Protein Equilibrium Dialysis and Parallel Artificial Membrane Permeability assays with bovine serum and biological matrices.
- Operated and troubleshot HPLC-UV/Vis and HPLC-MS systems for drug-protein interaction analysis using chromeleon.

Graduate Teaching Assistant

Sam Houston State University, TX, USA *August 2023 – May 2025*

- Led undergraduate labs in General, Inorganic, Biochemistry, Instrumental, and Integrated Chemistry.
- Trained students in titration, calorimetry, chromatography, UV-Vis assays, enzyme kinetics, and safe lab practices.
- Provided instruction on JASCO UV-Vis, Hitachi Fluorescence, Varian AA, and GC-FID instruments with focus on method setup and data interpretation**.**

Intern Biochemist

Birla Institute of Scientific Research, Rajasthan, India *Summer 2023*

- Hands-on training in microbiology, molecular biology, protein extraction, and bioinformatics.

Intern Chemist

Anant Pharmaceuticals, Haryana, India *Summer 2022*

- Performed QC analysis including dosage, coating assessments, and sample integrity testing.

SCIENTIFIC PRESENTATIONS

Esterellas Marquina, F. J.; **Khurana, A.**; Gonzalez, K.; Ortiz, S; Petrikovics, I.; Thompson, D.E. Gathering a basis set of UV-vis spectra in support of future studies of the interations between dimethyl trisulfide and albumin. 18th Annual Undergraduate Research Symposium, Sam Houston State University, Huntsville, Texas. April 2025.

Khurana, A.; Gonzalez, K.; Petrikovics, I; Thompson, D.E. *In vitro* investigation of dimethyl disulfide, a metabolite of the cyanide antidote candidate dimethyl trisulfide. *American Chemical Society Southwest Regional Conference,* Waco, Texas. October 2024 (Paper ID - 4165425)

Khurana, A.; Jordan, K; Perez, A.; Petrikovics, I; Thompson, D.E. Disambiguating the binding equilibria of volatile and nonvolatile compounds to protein. *American Chemical Society National Conference*, Denver, Colorado. August 2024 (Paper ID – 4108246)

Khurana, A.; Esterellas Marquina, F. J.; Villanueva, T.; Vazquez, K.; Rafeedie, N.; Langley, M.; Lopez, R.; Gray, K.; Black, K.; Herath, D.; Jordan, K.; Thompson, D.E.; Petrikovics, I. Protein binding studies for the cyanide antidote candidate dimethyl trisulfide. *127th Annual Meeting of Texas Academy of Science*, Odessa, Texas. March 2024

SERVICE

International Students Organization, *Event Coordinator*
SHSU, Huntsville, TX, USA *2023-2024*

Rotaract Club of Samaritans, *Joint Secretary; International Services Director*
Chandigarh, India *2021-2023*

National Cadet Corps, *Cadet (Certificate A, B and C holder)*
PHHP&C Directorate, Ministry of Defense, Government of India *2016-2022*